COHERENT QUADRANT

# Drone Batteries Industry Analysis

**India**

119 Manufacturer scored on Product Capability (X) and Business Capability (Y)

## Market Classification

**Market**

Drone Batteries Industry Analysis

**Market Type**

The development, production, and supply of high-performance energy storage units and power management systems specifically engineered for unmanned aerial vehicles (UAVs).

**Provider Categories in this Market**

1. Manufacturer

Manufacturer was chosen because the defining offering of this market is a physical energy storage device produced under a company's own name, making physical hardware production the primary basis of competition.

| 119 | 19 | 20 | 4 |
|---|---|---|---|
| Companies assessed | Companies on the Coherent Quadrant | Scoring parameters | Quadrants |

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## Market Scoring Parameters

**X — Product Capability**

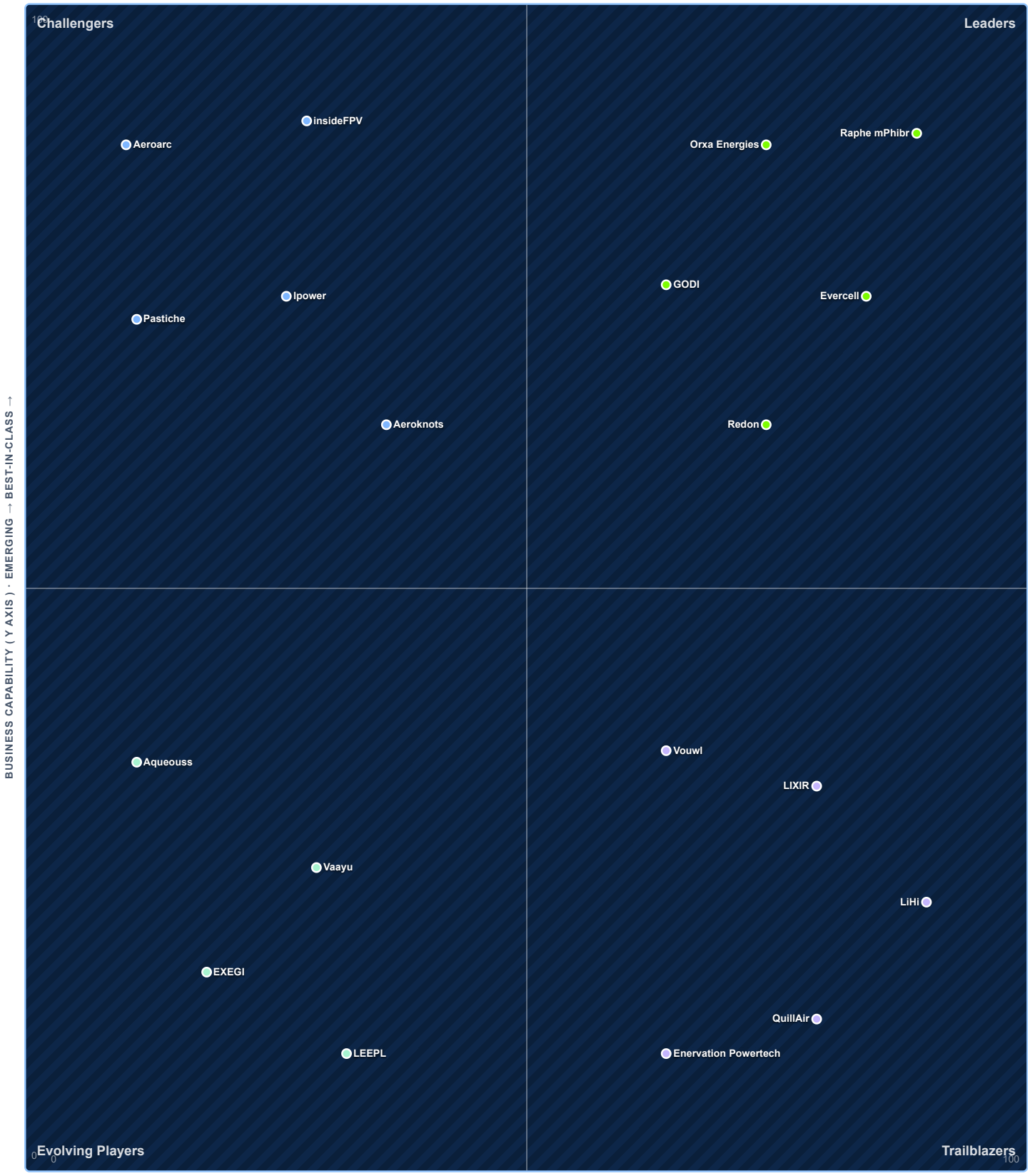
1. **Energy Density & Flight Time**
2. **Battery Chemistry & Safety**
3. **Customization & Form Factor Flexibility**
4. **Cycle Life & Durability**
5. **Charging Speed & Smart BMS**

**Y — Business Capability**

1. **OEM Partnerships & Contract Wins**
2. **Revenue & Growth Rate**
3. **Manufacturing Capacity & Localization**
4. **Distribution & After-Sales Network**
5. **Funding & Strategic Investments**

## Quadrant Positioning

(19 on chart · 119 in total)




**Countries on chart (by quadrant)**

**Leaders**: India×5 · **Challengers**: India×5 · **Trailblazers**: India×5 · **Evolving Players**: India×4

## What Each Quadrant Means

**Leaders** **Integrated Market Leaders**

Best-in-class Product Capability, Best-in-class Business Capability

Companies demonstrating strong product capabilities and robust business execution. They combine mature and differentiated offerings with established market presence, customer reach, commercial scale, and effective growth strategies, positioning them strongly for sustained market leadership.

**Challengers** **Scale-Driven Challengers**

Emerging Product Capability, Best-in-class Business Capability

Companies with strong business presence and commercial capabilities, supported by established customer relationships, market reach, channels, or brand strength. However, their product portfolio may have relatively lower differentiation, breadth, maturity, or innovation compared with leading participants.

**Trailblazers** **Innovation-Driven Trailblazers**

Best-in-class Product Capability, Emerging Business Capability

Companies demonstrating strong product capabilities, innovation, or differentiated offerings, but with comparatively lower business scale or market penetration. Their growth potential depends on strengthening commercial execution, customer reach, partnerships, geographic presence, and overall market visibility.

**Emerging Players** **Foundation-Building Evolving**

Emerging Product Capability, Emerging Business Capability

Companies with developing product capabilities and relatively limited business presence. These participants may be at an earlier stage of market development, serve focused segments or geographies, and have opportunities to strengthen both their product proposition and commercial footprint.

## Top 19 Companies

(plotted on the Quadrant Positioning chart)

| BRAND | COMPANY | HQ | ROLE | QUADRANT | X (PRODUCT CAPABILITY) | Y (BUSINESS CAPABILITY) | OVERALL SCORE |
|---|---|---|---|---|---|---|---|
| Orxa Energies | Orxa Energies | Bangalore, India | Manufacturer | Leaders | 90 | 96 | 93 |
| Evercell | Bharat Skytech Private Limited | New Delhi, India | Manufacturer | Leaders | 93 | 92 | 92 |
| Raphe mPhibr | Raphe mPhibr Private Limited | Noida, India | Manufacturer | Leaders | 91 | 94 | 92 |
| Redon | Redon India | Noida, India | Manufacturer | Leaders | 86 | 88 | 87 |
| GODI | Godi India Private Limited | Hyderabad, India | Manufacturer | Leaders | 84 | 90 | 87 |
| insideFPV | InsideFPV Ventures Private Limited | Surat, India | Manufacturer | Challengers | 80 | 97 | 88 |
| Aeroknots | Aeroknots | Lucknow, India | Manufacturer | Challengers | 84 | 86 | 85 |
| Ipower | IPower Batteries Private Limited | Sonipat, India | Manufacturer | Challengers | 78 | 90 | 84 |
| Aeroarc | Aeroarc Private Limited | South Delhi, India | Manufacturer | Challengers | 74 | 93 | 84 |
| Pastiche | Pastiche Energy Solutions Private Limited | Panchkula, India | Manufacturer | Challengers | 75 | 91 | 83 |
| LiHi | AKSI AI Aerospace and Agri Technology Ventures Private Limited | Hyderabad, India | Manufacturer | Trailblazers | 91 | 83 | 87 |
| QuillAir | QuillAir Aerospace and Engineering Private Limited | Kolkata, India | Manufacturer | Trailblazers | 90 | 82 | 86 |
| Vouwl | Techlanz Private Limited | Bangalore, India | Manufacturer | Trailblazers | 87 | 86 | 86 |
| LIXIR | Lixir Energy Private Limited | Pune, India | Manufacturer | Trailblazers | 88 | 84 | 86 |
| Enervation Powertech | Enervation Powertech Private Limited | Noida, India | Manufacturer | Trailblazers | 85 | 75 | 80 |
| Vaayu | Exodrone Systems Private Limited | Hyderabad, India | Manufacturer | Evolving Players | 82 | 85 | 84 |
| EXEGI | RCRS Innovations Limited | Noida, India | Manufacturer | Evolving Players | 82 | 83 | 82 |
| LEEPL | Lithium Energy Experts Pvt. Ltd. | Noida, India | Manufacturer | Evolving Players | 83 | 79 | 81 |
| Aqueouss | Aqueouss | New Delhi, India | Manufacturer | Evolving Players | 72 | 86 | 79 |

## Other Noticeable Players

(100 companies)

| BRAND | COMPANY | HQ | ROLE | RELATIVE POSITIONING | STRENGTH |
|---|---|---|---|---|---|
| mPower | Magnipower Technology Pvt. Ltd. | Chennai, India | Manufacturer | Leaders | |
| Adiabatic | Adiabatic Technologies Private Limited | Ahmednagar, India | Manufacturer | Leaders | |
| Dreamfly Innovations | Dreamfly Innovations Private Limited | Bengaluru, India | Manufacturer | Leaders | |
| Future Hi-Tech | Future Hi-Tech Batteries Limited | Mohali, India | Manufacturer | Leaders | |
| HBL | HBL Engineering Limited | Hyderabad, India | Manufacturer | Leaders | |
| Nexfly | Leolus Energy Private Limited | Bengaluru, India | Manufacturer | Leaders | |
| Qmax Ion | Qmax Ion Private Limited | Chennai, India | Manufacturer | Leaders | |
| Revogreen | Revogreen Technologies Pvt. Ltd. | Pune, India | Manufacturer | Leaders | |
| Trydan Tech | Trydan Tech | Bengaluru, India | Manufacturer | Leaders | |
| G-Power | G-Power Solutions | Chennai, India | Manufacturer | Leaders | |
| Garuda Aerospace | Garuda Aerospace Private Limited | Chennai, India | Manufacturer | Leaders | |
| GenX | Dazzle Robotics Pvt Ltd | Gandhinagar, India | Manufacturer | Leaders | |
| Ascend Powerpacks | Ascend Powerpacks Pvt. Ltd. | Pune, India | Manufacturer | Leaders | |
| IoTech | Bharath Components | Bengaluru, India | Manufacturer | Leaders | |
| HUB-E | HUB-E | Bengaluru, India | Manufacturer | Leaders | |
| HES Battery | Hes Battery Experts Private Limited | Noida, India | Manufacturer | Leaders | |
| Trontek | Trontek Electronics Pvt. Ltd. | Delhi, India | Manufacturer | Leaders | |
| GenX UAV Power | Robokits India | Gandhinagar, India | Manufacturer | Leaders | |
| KITES Power | Kites Aerospace | Thane, India | Manufacturer | Leaders | |
| LionBattery | Lion Battery | Umargam, India | Manufacturer | Leaders | |
| Pro-Range | Macfos Limited | Pune, India | Manufacturer | Leaders | |
| Voltherm Technologies | Voltherm Technologies LLP | Ahmedabad, India | Manufacturer | Leaders | |
| Flyotec | FlyoTec Innovation | Kolkata, India | Manufacturer | Leaders | |
| Voltherm | Voltherm Innovation Private Limited | Gandhinagar, India | Manufacturer | Leaders | |
| AlgoFET | AlgoFET | Bengaluru, India | Manufacturer | Leaders | |
| axl everfly | Axldrone | Kochi, India | Manufacturer | Leaders | |
| Astra | Hastin Energy Pvt. Ltd. | Bengaluru, India | Manufacturer | Leaders | |
| Micronix | Micronix | Bengaluru, India | Manufacturer | Leaders | |
| ThunderVolt | Syfero Synergetics Drone Solutions Private Limited | Coimbatore, India | Manufacturer | Leaders | |
| Shivon | Shivon Energy Private Limited | Bengaluru, India | Manufacturer | Leaders | |
| Sytiqhub EV | Sytiqhub EV Solutions Pvt. Ltd. | Vijayawada, India | Manufacturer | Leaders | |
| Faradic | Espacio Aeronext Private Limited | Rajpur Sonarpur, India | Manufacturer | Leaders | |
| ATC Batteries | ATC Energies Systems Ltd | Noida, India | Manufacturer | Leaders | |
| Equbz | Equbz Lithium Battery Solutions | Coimbatore, India | Manufacturer | Leaders | |
| Jackvolt | Shaft Energies Private Limited | Noida, India | Manufacturer | Leaders | |
| Artek | Artek Energy Pvt Ltd | Noida, India | Manufacturer | Leaders | |

| | | | | | |
|---|---|---|---|---|---|
| Euclion Energy | Euclion Energy Private Limited | Noida, India | Manufacturer | Leaders | |
| EV Battery Solutions | EV Battery Solutions | Bengaluru, India | Manufacturer | Leaders | |
| Eco Vehicle | Eco Vehicle | Kolkata, India | Manufacturer | Leaders | |
| Ampton Energy | Ampton Energy Private Limited | Chennai, India | Manufacturer | Challengers | |
| Komaki | KLB Komaki Pvt. Ltd. | New Delhi, India | Manufacturer | Challengers | |
| Mobilus | Sys-Net Technologies | Vadodara, India | Manufacturer | Challengers | |
| Mavdrones | Maverick Drones & Technologies Pvt. Ltd. | Hyderabad, India | Manufacturer | Challengers | |
| Greenmax | Green Max Technology | Bhopal, India | Manufacturer | Challengers | |
| Pinnacle | Pinnacle Power Systems | Coimbatore, India | Manufacturer | Trailblazers | |
| Neuron Energy | Neuron Energy Private Limited | Mumbai, India | Manufacturer | Challengers | |
| HeatRay Solar | HeatRay Solar Pvt Ltd | Indore, India | Manufacturer | Trailblazers | |
| Hison Energies | Hison Energies Pvt. Ltd. | Vadodara, India | Manufacturer | Trailblazers | |
| Cellex | Cellex Battery Systems Pvt. Ltd. | Coimbatore, India | Manufacturer | Trailblazers | |
| Devika Teletronic | Devika Teletronic | New Delhi, India | Manufacturer | Trailblazers | |
| MEB Energy | MEB Energy LLP | Surat, India | Manufacturer | Trailblazers | |
| Vimaana Aerospace Technologies | Vimaana Aerospace Technologies LLP | Greater Noida, India | Manufacturer | Trailblazers | |
| Caryvolt | Caryaire Equipments India Pvt. Ltd. | Noida, India | Manufacturer | Challengers | |
| One Origin India | One Origin India | Noida, India | Manufacturer | Trailblazers | |
| Seashield Technology | Seashield Technology | Haridwar, India | Manufacturer | Challengers | |
| Elbotics | Elbotics Batteries | Jaipur, India | Manufacturer | Evolving Players | |
| Advantex | Advantex Battery LLP | Kolkata, India | Manufacturer | Trailblazers | |
| Batteryinc | Batteryin Private Limited | Bengaluru, India | Manufacturer | Trailblazers | |
| Eurovolta | Eurovolta Innovations Pvt Ltd | Gurugram, India | Manufacturer | Trailblazers | |
| Hyreach | Hyreach Technologies | Chennai, India | Manufacturer | Trailblazers | |
| Vivitron | Vivitron Energy Pvt. Ltd. | Bengaluru, India | Manufacturer | Trailblazers | |
| Aerodrone Robotics | Aerodrone Robotics Private Limited | Gurugram, India | Manufacturer | Evolving Players | |
| Skypoint Industries | Skypoint Industries | Chennai, India | Manufacturer | Evolving Players | |
| Amptek | Amptek Batteries | Gurugram, India | Manufacturer | Evolving Players | |
| Seirei Electronics | Seirei Electronics Services Private Limited | Noida, India | Manufacturer | Evolving Players | |
| Techsolve | Techsolve | Bengaluru, India | Manufacturer | Evolving Players | |
| Boson | EWC Recycling Private Limited | Chennai, India | Manufacturer | Evolving Players | |
| Battrixx | Kabra Extrusiontechnik Limited | Mumbai, Country | Manufacturer | Challengers | |
| Scion Energy Storage | Scion Energy Storage | Pune, India | Manufacturer | Evolving Players | |
| Kisan Tools | Syagro Kisan Tools Private Limited | Indore, India | Manufacturer | Challengers | |
| Vayukah | Vayukah Drone Systems And Services Pvt. Ltd. | Vijayawada, India | Manufacturer | Evolving Players | |
| Vishvatronics Urja | Vishvatronics Urja Private Limited | Noida, India | Manufacturer | Evolving Players | |
| Bentork | Bentork Industries LLP | Pune, India | Manufacturer | Evolving Players | |
| Enlitso | Enlitso Energy Private Limited | New Delhi, India | Manufacturer | Evolving Players | |
| Pantagone Technologies | Pantagone Technologies India Pvt. Ltd. | Bhopal, India | Manufacturer | Evolving Players | |

| | | | | | |
|---|---|---|---|---|---|
| Aviocian Technologies | Aviocian Technologies Private Limited | Greater Noida, India | Manufacturer | Trailblazers | |
| Eteily | Eteily Technologies India Pvt. Ltd. | Bhopal, India | Manufacturer | Evolving Players | |
| LRSA | LRSA Technologies Pvt. Ltd. | Delhi, India | Manufacturer | Evolving Players | |
| VoltBee | VoltBee | Kolkata, India | Manufacturer | Evolving Players | |
| Bluewatt Technology | Bluewatt Technology | Bhopal, India | Manufacturer | Evolving Players | |
| Mikrobotix | Mikrobotix | Karimnagar, India | Manufacturer | Evolving Players | |
| Varsha Engineering | Varsha Engineering | Noida, India | Manufacturer | Evolving Players | |
| Voltech Power Solutions | Voltech Power Solutions | Hyderabad, India | Manufacturer | Evolving Players | |
| Autorev Engineering | Autorev Engineering Private Limited | Bhopal, India | Manufacturer | Challengers | |
| Robocraft | Robocraft Impex | Ahmedabad, India | Manufacturer | Challengers | |
| RGBSN Energy | RGBSN Energy Pvt. Ltd. | Noida, India | Manufacturer | Evolving Players | |
| Energenic | Energenic Batteries LLP | Faridabad, India | Manufacturer | Challengers | |
| Eclectech | Eclectech Power Packs LLP | Chennai, India | Manufacturer | Evolving Players | |
| Cygni | Cygni Energy Private Limited | Hyderabad, India | Manufacturer | Evolving Players | |
| High Energy | High Energy Batteries (India) Limited | Chennai, India | Manufacturer | Evolving Players | |
| Maxvolt Energy | Maxvolt Energy Industries Limited | Ghaziabad, India | Manufacturer | Evolving Players | |
| Lithience Energy | Lithience Energy | Ghaziabad, India | Manufacturer | Evolving Players | |
| Inverted | Inverted Energy Pvt Ltd | New Delhi, India | Manufacturer | Evolving Players | |
| Randal Enterprise | Randal Enterprise | Surat, India | Manufacturer | Evolving Players | |
| Jalaram Lithium Battery | Jalaram Lithium Battery India Private Limited | Ahmedabad, India | Manufacturer | Evolving Players | |
| a2zRC | RCHobbyTech Solutions Pvt. Ltd. | Kolkata, India | Manufacturer | Evolving Players | |
| VEMAC | Vibrant Engineering Mechanics & Automation Controls | Ahmedabad, India | Manufacturer | Evolving Players | |
| Bhishma | Bhishma U.A.V Technologies | Hyderabad, India | Manufacturer | Evolving Players | |
| Li3 | Sanvaru Technology Limited | Sonipat, India | Manufacturer | Evolving Players | |
| Efficell | Efficell Power Solutions | Coimbatore, India | Manufacturer | Evolving Players | |

# Parameter Detail by Company

(Product Capability and Business Capability evidence for the 19 companies on the chart)

## Orxa Energies ● Leaders

X **90** Y **96** Overall **93**

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 91/100

> Although Orxa Energies demonstrates practical battery integration and regulatory certification, the absence of granular metrics on energy density and discharge rates limits verification to general company claims. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Orxa Energies manufactures high-power, high-density, and compact battery packs tailored for medium-to-heavy drone categories. Based on primary research, the company has not disclosed precise specifications such as Wh/kg, C-rate, or continuous discharge rates.
- The company achieved a milestone by receiving type-certification in India for a medium-class drone battery pack and collaborated with UrbanMatrix for civil UAV applications.

#### Cycle Life & Durability — scored 94/100

> The company has a partial capability through type certification and BMS enhancements, but specific environmental and cycle-life test data for UAV applications are not fully detailed in public records.

SUPPORTING POINTS

- Orxa Energies' battery packs have achieved type certification in India for the medium-class drone category, and the company has tested its battery architecture with custom BMS layouts indicating improved cycle life under high-load usage.
- Specific technical ratings for drone battery cycle counts, vibration resistance, and humidity performance remain limited to broader public road testing metrics from other vehicle platforms.

#### Safety & Thermal Management — scored 85/100

> The company explicitly highlights its internal R&D and proprietary thermal management capability for heavy UAVs, but the lack of publicly verifiable, independent third-party safety certifications limits its evaluation to self-reported evidence. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Orxa Energies manufactures high-performance lithium-ion battery packs for heavy drones in aerospace, defense, and agricultural sectors, emphasizing compact size, high energy density, and custom-engineered thermal management systems.
- Based on primary research, the company has not disclosed information for this parameter. No third-party verified safety certifications such as UN38.3 or IEC 62133 specifically for their UAV drone battery line could be publicly located.

#### Customization & Form Factor Flexibility — scored 100/100

> While Orxa Energies has self-reported capabilities in custom engineering high-power battery architectures and proprietary BMS solutions for heavy drones, specific details on modular product catalogs or rapid prototyping services for drone customers lack extensive public third-party validation.

SUPPORTING POINTS

- Orxa Energies specializes in designing and manufacturing high-performance Lithium-ion battery packs tailored for heavy drones in aerospace and defense applications.
- The company leverages proprietary Battery Management Systems (BMS), thermal management configurations, and deep engineering design expertise to fit unique B2B aerospace requirements and demanding drone mission profiles.

#### R&D & Innovation in Battery Chemistry — scored 78/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. The lack of public evidence of internal development of advanced chemical formulations limits this assessment.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed evidence regarding proprietary investment in next-generation advanced cell chemistries such as solid-state or lithium-sulfur by Orxa Energies.

## Y — BUSINESS CAPABILITY

### OEM Partnerships & Customer Contracts — scored 100/100

> Secondary research confirms a specific named drone OEM partnership (UrbanMatrix) along with an explicit commercial contract valuation (₹20 Crore), but based on primary research, the company has not disclosed named defense client customer lists or individual case studies, given confidentiality protocols surrounding its aerospace and defense segments.

SUPPORTING POINTS

- The company has a strategic research and development alliance estimated at ₹20 Crore with Bangalore-based drone manufacturer UrbanMatrix to co-develop high-power, high-capacity lithium-ion battery packs for medium-to-heavy class drones targeting the global industrial and civil export markets.

### Production Capacity & Scalability — scored 96/100

> While general factory expansion and assembly setups are verified, specific drone-specific manufacturing output metrics and scaling validations for UAV orders are unverified, leaving this a self-reported, partial capability. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Orxa Energies opened a new 1-acre manufacturing and battery assembly facility in the Bommasandra Industrial Area, Bengaluru, expanding its overall manufacturing capacity ten-fold.
- The facility features dedicated battery assembly lines and an annual production capacity geared toward its electric vehicle applications.

### Revenue Growth & Financial Health — scored 91/100

> Secondary research confirms the external venture capital funding status and milestone B2B partnership, but complete verified fiscal year revenue sheets and precise net profitability metrics are lacking to independently confirm sustained financial growth.

SUPPORTING POINTS

- In December 2023, Orxa Energies formed an R&D and manufacturing alliance with drone technology company UrbanMatrix valued at ₹20 Crore (INR 200 million) to develop type-certified battery packs for civil applications.
- According to venture tracking data from Tracxn and PitchBook, Orxa Energies has raised a total of approximately $5.45 million to $5.73 million in external funding, including a Pre-Series A round led by Lectrix E-Vehicles.

### Geographic Reach & Distribution Network — scored 94/100

> Secondary research confirms the company's central manufacturing base in Bangalore and its active expansion strategy into export markets via a commercial drone partnership, but based on primary research, the company has not disclosed specific numbers of external regional dealers or active international countries served.

SUPPORTING POINTS

- A strategic research and development alliance valued at ₹20 Crore exists between UrbanMatrix and Orxa Energies, aimed explicitly at establishing an export business to address global civil industrial drone demands.
- The operation of a single consolidated 1-acre corporate research, development, and battery/vehicle assembly facility located at Bommasandra Industrial Area, Bangalore, India, from which they manufacture and locally distribute their high-performance drone battery packs.

### Certifications & Compliance for Defense/Aerospace — scored 98/100

> Secondary research confirmed Orxa's milestone BIS certification and defense platform integration, but the absence of publicly listed military standards or ISO 9001 registration leaves the company with a clear, verifiable capability rather than leading-edge aerospace compliance. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Orxa Energies achieved Bureau of Indian Standards (BIS) certification in the medium to heavy drone class in India, as documented by UrbanMatrix.
- Orxa Energies' proprietary battery pack technology has been applied on defense and aerospace platforms, with high-performance aerospace packs undergoing formal aerospace certification protocols according to Unreasonable Group.

## Bharat Skytech Private Limited Evercell ● Leaders X 93 Y 92 Overall 92

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 100/100

> The rating rests on the supporting points listed for this parameter. Based on primary research, the company has not disclosed further independently verified details.

SUPPORTING POINTS

- The company's upcoming Evercell product line expansion includes High-Voltage (HV) LiPo batteries, Semi-Solid batteries, and Semi-Solid Smart Batteries spanning capacities from 16000 mAh up to 30000 mAh to optimize flight endurance and capacity.

#### Cycle Life & Durability — scored 96/100

> The company provides self-reported technical specifications claiming a 500+ charge cycle life and general environmental resilience for its specialized Evercell line, but lacks independent, third-party verified test reports or standardized laboratory certificates on public record to validate these endurance claims. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Bharat Skytech Private Limited self-reports that its "Make in India" Evercell 6S 22.2V 22000mAh LiPo drone battery features an anti-swelling cell construction engineered to hold pack geometry and sustain 500+ charge cycles across intensive daily utilization.
- The company claims that the Evercell range utilizes robust cell construction and is built to deliver stable performance across the distinct heat, humidity, dust, and vibration conditions of Indian field and industrial environments.

#### Safety & Thermal Management — scored 92/100

> The company has a documented capability through mandatory national safety certifications (BIS) and standard BMS thermal protections, rather than carrying leading-edge global certifications such as UN38.3 or IEC 62133.

SUPPORTING POINTS

- Bharat Skytech's Evercell drone battery line is manufactured under the Make in India initiative with official Bureau of Indian Standards (BIS) certification, validating compliance with safety benchmarks against overcharging, short circuits, and thermal runaway.
- The Evercell HV LiPo Smart Battery features an integrated Battery Management System (BMS) that provides real-time individual cell monitoring, automatic balancing, and a hardware-level short-circuit and thermal overload safety cutoff.

#### Customization & Form Factor Flexibility — scored 79/100

> The company primarily markets a set inventory of pre-configured, standardized 6S battery configurations tailored to common 10L-16L agricultural platforms rather than advertising on-demand physical custom sizing, bespoke form-factor prototyping, or client-tailored modular design services for non-standard UAV hulls.

SUPPORTING POINTS

- While Bharat Skytech operates as a component supplier enabling users to custom-assemble or upgrade agricultural drones, its actual battery catalog lists pre-configured off-the-shelf options from its own Evercell line and distributed brands like Herewin.

#### R&D & Innovation in Battery Chemistry — scored 100/100

> Bharat Skytech has demonstrated capability by transitioning from an online component distributor into local battery manufacturing with its Evercell line, introducing advanced semi-solid chemistry alongside smart BMS tracking.

SUPPORTING POINTS

- Bharat Skytech Private Limited has launched its own "Make in India" brand called Evercell, which includes conventional Lithium Polymer (LiPo), High-Voltage Lithium Polymer (HV LiPo), and intelligent smart batteries integrated with Battery Management Systems (BMS).
- The company introduces "Evercell Semi-Solid Batteries" on its platform, explicitly highlighting the use of advanced semi-solid electrolyte chemistry to achieve higher energy densities and improved thermal stability compared to liquid electrolyte packs.

## Y — BUSINESS CAPABILITY

### OEM Partnerships & Customer Contracts — scored 97/100

> Bharat Skytech has documented capabilities through a major manufacturing co-branding partnership with EFT and established distribution networks with global component giants, but lacks publicly detailed long-term supply contracts or case studies verifying market-leading defense or commercial drone OEM battery adoption. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- In March 2026, Bharat Skytech Private Limited established a strategic co-brand partnership named "EFT BHARAT" with global UAV technology developer EFT to localise manufacturing, distribution, and service of the E610P agricultural drone in India.
- Bharat Skytech serves as an authorized distribution partner for major global drone component and battery manufacturers, including Hobbywing, SIYI, JIYI, EFT, Tattu, and Herewin.

### Production Capacity & Scalability — scored 93/100

> The company shows evidence of establishing domestic manufacturing lines and launching a dedicated battery brand, but the lack of specific, third-party verified annual battery output data leaves this evidence partial and self-reported. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Bharat Skytech established a dedicated manufacturing facility to produce localized "Make in India" drone components under its brand, Evercell, which includes high-performance drone batteries and smart charging systems.
- Parent company Thakral Corporation announced the setup of in-house production lines for enterprise drone parts. Based on primary research, the company has not disclosed specific battery manufacturing output metrics such as MWh/year.

### Revenue Growth & Financial Health — scored 97/100

> The company has third-party verified commercial capability, with strong upward revenue growth and solid corporate backing, a clear and verifiable strength.

SUPPORTING POINTS

- According to registry profiles, Bharat Skytech Private Limited is backed by Singapore's Thakral Corporation and generated net sales revenues between ₹50 Cr and ₹100 Cr for the financial year ending March 31, 2025, reflecting a 47.44% year-over-year net sales growth rate.
- The company operates its Evercell brand, producing LiPo and semi-solid smart batteries for UAVs from its dedicated Indian manufacturing facility.

### Geographic Reach & Distribution Network — scored 92/100

> The company shows partial, self-reported geographic reach, offering pan-India direct-to-customer logistics and executing distributor enrollment programs from a single centralized hub, though it lacks a verified network of permanent third-party physical dealers or international export sales.

SUPPORTING POINTS

- Bharat Skytech Private Limited operates a centralized domestic fulfillment hub out of its main corporate/warehousing offices in Gurugram, Haryana, offering pan-India e-commerce shipping and fast delivery to drone operators across the country.
- The company is actively inviting local dealers and regional distributors through direct commercial onboarding programs to scale its "Evercell" drone battery brand across Indian agricultural regions.

### Certifications & Compliance for Defense/Aerospace — scored 83/100

> The company has demonstrated operational and regulatory capability for commercial and agricultural drone components (such as WPC equipment approvals) and assists with DGCA standards, but it lacks documented, third-party verified aerospace or defense-grade military compliance certificates (like MIL-STD) for its specific drone battery lines. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Bharat Skytech Private Limited holds an official authorization certificate for its "WPC Approved B12 Bharat Ground Control System" and offers compliance guidance helping clients align with Directorate General of Civil Aviation (DGCA) standard requirements for commercial and agricultural drones.
- Based on primary research, the company has not disclosed aerospace-specific or military-grade testing credentials such as MIL-STD or specialized Indian defense procurement standard certifications for their power units.

## Raphe mPhibr Private Limited Raphe mPhibr ● Leaders X 91 Y 94 Overall 92

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 100/100

> The company manufactures its own proprietary battery packs in-house with distinct C-rate specifications available, a demonstrated capability, though specific gravimetric energy density (Wh/kg) verification remains absent from public military-grade documentation.

SUPPORTING POINTS

- Raphe mPhibr Private Limited manufactures all of its own drone components in-house, explicitly producing its own batteries and power infrastructure domestically in India to maintain an independent supply chain.

#### Cycle Life & Durability — scored 77/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. Although the company indigenously develops its own UAV batteries and its drones are operationally deployed by the Indian Armed Forces in harsh terrains, specific metrics on battery cycle life ratings, shelf life, or formal environmental test results remain entirely undisclosed in public sources.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

#### Safety & Thermal Management — scored 84/100

> The company shows partial, self-reported capabilities in providing thermal management features and heat-safe drone battery packs. Based on primary research, the company has not disclosed specific third-party safety certifications (such as UN38.3 or IEC 62133).

SUPPORTING POINTS

- Raphe mPhibr designs and manufactures custom battery packs, proprietary battery management systems (BMS), and thermal/heat-safe pack solutions specifically engineered to be non-explosive and thermally stable to prevent overheating.

#### Customization & Form Factor Flexibility — scored 100/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Raphe mPhibr manufactures almost all drone subsystems domestically from scratch—including flight controllers, autopilots, inertial navigation, and batteries—utilizing fully vertically integrated in-house electronics R&D and PCB A manufacturing.
- The company leverages an expansive advanced manufacturing infrastructure featuring over 80 3D printers (including the largest fleet of Formlabs SLA printers in India) and Philips additive hybrid technology to achieve high flexibility in custom structural part rapid prototyping and production.

#### R&D & Innovation in Battery Chemistry — scored 94/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Raphe mPhibr manufactures its own custom drone battery systems and packs entirely in-house at its vertically integrated manufacturing facilities to ensure complete supply chain independence from external components.
- In interviews, the company's CEO highlights their strategic trade-offs regarding battery technology limitations, stating that while they work intimately on battery systems, their advanced long-endurance heavy-lift UAV solutions heavily prioritize internal combustion engines (ICE) because battery energy densities cannot yet compete with fossil fuels for extended flight durations.

### Y — BUSINESS CAPABILITY

#### OEM Partnerships & Customer Contracts — scored 97/100

> The company has verifiable vertical integration and defense deployment, but lacks external OEM customer contracts for standalone battery supply.

SUPPORTING POINTS

- Raphe mPhibr manufactures in-house custom drone battery packs and battery management systems to power its logistics and swarm drone contracts with the Indian Army.
- Public evidence is absent regarding merchant supply agreements or standalone battery sales to external commercial drone OEMs.

### Production Capacity & Scalability — scored 94/100

> While the company has large-scale, vertically integrated manufacturing capacity for full UAV assemblies (including in-house drone battery fabrication), there is no specific public breakdown of its standalone battery manufacturing capacity in terms of annual MWh output or dedicated battery line metrics.

SUPPORTING POINTS

- Raphe mPhibr indigenously produces its own drone batteries alongside flight controllers and other structural materials at its integrated research and production hub in Noida.
- Following a $100 million funding round in mid-2025, the company expanded its manufacturing footprint from a 100,000-square-foot facility to a 650,000-square-foot campus to scale overall production to over 2,000–2,500 UAVs annually.

### Revenue Growth & Financial Health — scored 100/100

> The company shows exceptionally high revenue growth, solid profitability, and substantial third-party private equity backing, a clear and verifiable capability in the Indian defense tech and UAV sub-system sectors.

SUPPORTING POINTS

- Raphe mPhibr Private Limited reported an operating revenue range of ₹100 crore to ₹500 crore for the financial year ending March 31, 2025, demonstrating significant year-on-year revenue growth alongside sustained profitability.
- In June 2025, the company secured a $100 million Series B equity funding round led by General Catalyst, bringing its total capital raised to $145 million at a valuation of nearly $900 million to expand its domestic UAV and component manufacturing.

### Geographic Reach & Distribution Network — scored 81/100

> The company uses a heavily centralized, vertically integrated manufacturing setup within a single campus in India to sell directly to government or defense clients, leaving weak or unproven evidence of a broader regional distributor/dealer network or export distribution footprint. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed external retail dealers, regional service centers, or international export distribution agreements.

### Certifications & Compliance for Defense/Aerospace — scored 100/100

> The company has demonstrated capability through heavy-lift defense deployment and active compliance simulation frameworks, although its specific third-party drone battery certifications (like MIL-STD listings) are not fully itemized publicly.

SUPPORTING POINTS

- Public records confirm that Raphe mPhibr uses advanced simulation tools via the Dassault Systèmes 3DEXPERIENCE platform (specifically SIMULIA) to respond to electromagnetic compliance, interference, and aerospace certification efforts.
- The company works tightly with Indian defense procurement norms, supplying combat-ready high-altitude logistics, swarm, and precision drones directly to the Indian Armed Forces, while hosting formal delegations from the National Defence College to showcase its indigenous testing infrastructure.

## Redon India Redon ● Leaders X 86 Y 88 Overall 87

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 85/100

> While secondary research confirms that Redon India targets the commercial and agricultural UAV market with custom lithium-ion battery packs, specific engineering metrics like Wh/kg, C-rate, and continuous power output remain unproven and self-reported, without published technical data sheets or third-party laboratory verification.

SUPPORTING POINTS

- Redon India officially lists "Drone Battery" under its main product groups alongside EV and Telecom batteries on its primary web domain, stating general utilization of customized lithium-ion technology to deliver higher energy density and faster charging.

#### Cycle Life & Durability — scored 94/100

> Although Redon India states it engineers customized packs for commercial/agricultural drones and demonstrates testing competency via NATRAX automotive certifications, specific third-party test data on UAV-specific flight cycle life, temperature swings, and humidity levels remain self-reported or unvouched.

SUPPORTING POINTS

- Redon India officially reports generalized cycle life parameters for its advanced lithium-ion technologies on its platform, noting that their lithium cells generally exhibit a lifecycle of 3000-5000 cycles under optimized high-discharge rates up to 10C alongside a minimal self-discharge rate of less than 3% per month, as detailed in the Redon India Social Releases.

#### Safety & Thermal Management — scored 85/100

> While built-in protective BMS circuits and thermal monitoring were verified through documentation, explicit third-party international safety certifications specific to the drone battery line could not be authenticated, leaving this a partial capability. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- The company operates a factory in Greater Noida where customized lithium battery packs undergo charge-discharge and safety verification tests, as reported on the Redon India Corporate Portal.

#### Customization & Form Factor Flexibility — scored 79/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Redon India outlines a standard 5-step engineering process on its official portal for building personalized, custom lithium energy solutions that begins with analyzing application-specific performance requirements and culminates in creating custom specifications, design blueprints, and initial prototype building/validation.

#### R&D & Innovation in Battery Chemistry — scored 86/100

> While production capabilities for lithium-ion battery packs are verified, there is a lack of public evidence on investments in advanced chemistries like solid-state or lithium-sulfur. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Redon India manufactures customized lithium-ion battery packs engineered for commercial and agricultural drone platforms.

### Y — BUSINESS CAPABILITY

#### OEM Partnerships & Customer Contracts — scored 80/100

SUPPORTING POINTS

- Redon India actively promotes custom manufacturing capabilities and lists "Drone Battery" alongside automotive and solar offerings on its dedicated OEM Partnerships Portal.
- General industry documentation confirms Redon India operates corporate and factory locations out of Noida while including customized drone battery packs among its main lithium-ion specialities on its LinkedIn Corporate Overview.

#### Production Capacity & Scalability — scored 98/100

> Secondary research confirms Redon India's facility dimensions and daily maximum product unit output through its self-reported trade exhibits and corporate literature, but based on primary research, the company has not disclosed the precise segment allocation or MWh/year metrics explicitly isolated for drone platforms.

SUPPORTING POINTS

- According to the official company profile featured on the IIEV Show Directory, Redon India operates a 12,000 square yard factory space capable of generating a production output of 2,500 battery packs per day.

### Revenue Growth & Financial Health — scored 87/100

> Secondary research confirms the company's baseline sales revenue scale and unfunded status using independent corporate databases, but based on primary research, the company has not disclosed specific profitability metrics and drone-exclusive segment growth figures.

SUPPORTING POINTS

- Redon Lithium Industries LLP generated annual sales revenue between ₹50 Cr and ₹100 Cr for the financial year ending March 31, 2025, according to data from a Tracxn Company Profile.
- The company has not raised any outside venture capital or institutional funding to date and operates as an unfunded active entity with a year-over-year revenue growth rate (CAGR) of -3% for the last fiscal year, per its financial summary on Tracxn.

### Geographic Reach & Distribution Network — scored 92/100

> Secondary research confirms Redon India's dual manufacturing footprints in Northern and Southern Indian regions alongside partial domestic retail distribution capability, but based on primary research, the company has not disclosed any documented cross-border distribution agreements or specific overseas export markets.

SUPPORTING POINTS

- Redon India maintains a corporate office at ATS Bouquet in Noida and an active manufacturing facility located at Sector Ecotech-1 Extension in Greater Noida, Uttar Pradesh, alongside a registered office and a production plant located in the Sompura Industrial Area (Dobaspet) near Bengaluru, Karnataka.
- The company leverages domestic distribution partnerships within India, including wholesaling and retail availability of its customized lithium-ion battery configurations via Harbacore dealership networks located in Kanpur, Uttar Pradesh.

### Certifications & Compliance for Defense/Aerospace — scored 82/100

> Secondary research confirms general manufacturing capability and basic Indian automotive/solar compliance. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Public product documentation confirms Redon India focuses primarily on automotive and consumer applications, holding certificates such as the AIS156 Certificate for its Redon 6039 battery pack issued by NATRAX, which ensures electrical safety for electric vehicles rather than defense or aerospace standards.
- While company landing pages on social platforms claim they serve sectors including Agriculture & Defense and custom-design drone batteries spanning 24V to 51.2V, official governance disclosures only list civilian certifications such as BIS Certification and being MNRE Approved for renewable energy and EVs.

## Godi India Private Limited GODI ● Leaders X 84 Y 90 Overall 87

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 70/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed evidence for drone or UAV-specific energy density (Wh/kg) or power output (C-rate) specifications.

#### Cycle Life & Durability — scored 100/100

> GODI India has clear, third-party verified testing (BIS/AIS) confirming a durability capability of over 2,000 cycles for its core lithium-ion cell technologies, but specific public evidence and environmental data explicitly mapping these metrics to a dedicated UAV/drone product line remain limited.

SUPPORTING POINTS

- GODI India's locally designed Lithium Iron Phosphate (LFP) cells have been certified by the Bureau of Indian Standards (BIS) and Automotive Industry Standards (AIS), demonstrating a reported service life surpassing 2,000 cycles alongside rigorous reliability and abuse testing.
- The company's technical specifications for its overall cell portfolio indicate that its energy and power cells are built to exceed 1,000 to 10,000 cycles depending on the specific form factor.

#### Safety & Thermal Management — scored 100/100

> GODI India demonstrates clear, third-party verified capability via TÜV and BIS certifications conforming to international cell safety standards (IEC 62133), making battery core safety one of its market-leading strengths.

SUPPORTING POINTS

- GODI India's Lithium Iron Phosphate (LFP) cells are certified by the Bureau of Indian Standards (BIS) and Automotive Industry Standards (AIS) after testing by TÜV to IS 16046 Part-2:2018 / IEC 62133-2:2017 standards.
- GODI India developed indigenous NMC811 21700 cylindrical lithium-ion cells that achieved BIS certification through third-party independent testing by TÜV.

#### Customization & Form Factor Flexibility — scored 65/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Godi India Private Limited's primary publicly stated capabilities focus on manufacturing standard certified cell formats, such as India's first Bureau of Indian Standards (BIS) certified 21700 cylindrical NMC811 lithium-ion cells.
- Based on primary research, the company has not disclosed further details.

#### R&D & Innovation in Battery Chemistry — scored 86/100

> While GODI India demonstrates advanced internal R&D capabilities in cell chemistries like silicon anodes and solid-state research, its innovation remains unproven or generalized regarding specific optimization for unmanned aerial vehicles (UAVs).

SUPPORTING POINTS

- GODI India has developed homegrown 5.2 Ah 21700 cylindrical lithium-ion cells with an energy density of 275 Wh/kg using advanced silicon anode technology, alongside an R&D pipeline that includes active research into sodium-ion and solid-state chemistries.
- The company is establishing a facility in Telangana dedicated to lithium-ion, sodium-ion, and related advanced battery technologies to cater broadly to sectors including defense and aerospace.

### Y — BUSINESS CAPABILITY

#### OEM Partnerships & Customer Contracts — scored 91/100

> The company shows verified custom battery design projects for strategic institutions like the DRDO and defense-adjacent aerospace MoUs, but there are no publicly named commercial drone manufacturer client lists or specific long-term supply contract values disclosed specifically for the UAV market.

SUPPORTING POINTS

- GODI India has established custom battery hardware projects for defense and aviation applications, including developing a prismatic cell battery pack and battery box for the Defence Research and Development Organisation (DRDO) and custom cylindrical battery packs for Russian Aviation (45Ah/26.9V), according to technical project portfolios documented by Karthik Gunnala (Mechanical Engineering, Godi India).

### Production Capacity & Scalability — scored 96/100

> Secondary research confirms the company's operational pilot line capabilities and its multi-gigawatt expansion plans via official state government agreements, but actual automated mass production specific to drone batteries remains unproven and largely self-reported at a pre-commercial scale.

SUPPORTING POINTS

- GODI India has established an operating lithium-ion cell pilot production line in Hyderabad and previously outlined a 100 MWh pilot manufacturing facility for its 5.0 - 5.2Ah 21700 cells.
- In January 2024, GODI India signed a Memorandum of Understanding (MoU) with the state government of Telangana to invest ₹8,000 crore ($962 million) to build a 12.5 GWh giga-scale cell manufacturing facility, targeting a 2.5 GWh assembly line in Phase 1 with planned expansion to 10 GWh in Phase 2.

### Revenue Growth & Financial Health — scored 89/100

> Secondary research confirms the company's funding status and net losses via public filings and market databases, but based on primary research, the company has not disclosed precise revenue metrics derived exclusively from drone-specific battery sales, as distinct from its broader portfolio,.

SUPPORTING POINTS

- Third-party financial reporting indicates that Godi India Private Limited recorded a 52.2% year-on-year revenue decline, posting a net loss of ₹34.6 Crore on total revenues of ₹77.9 Lakh for the financial year.

### Geographic Reach & Distribution Network — scored 81/100

> Secondary research confirms the company's localized pilot production and collaborative facility footprints across the South Indian region. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- A manufacturing and R&D presence limited domestically to a pilot production facility in Hyderabad and a public-private partnership agreement with the CSIR-CECRI institute to share an advanced cell manufacturing facility in Taramani, Chennai.
- The company announced a planned long-term expansion via a Memorandum of Understanding at the World Economic Forum to invest ₹8,000 crore ($962 million) in building a 12.5 GWh giga-scale battery cell manufacturing unit in Telangana.

### Certifications & Compliance for Defense/Aerospace — scored 91/100

> Secondary research confirms that the company holds third-party BIS certifications for its battery cells and actively targets the defense sector; however, based on primary research, the company has not disclosed direct evidence of drone-specific DGCA type certifications, MIL-STD defense qualification listings, or active compliance audits under Indian defense procurement norms.

SUPPORTING POINTS

- Manufacturing Today India reports that GODI India achieved Bureau of Indian Standards (BIS) certification for its homegrown Lithium Iron Phosphate (LFP) cells, following an earlier BIS certification for its NMC811 21700 cylindrical cells verified by third-party testing agency TUV, demonstrating standardized commercial component compliance within India.
- Battery-Tech Network lists "Defense and aerospace" as a primary target sector for GODI India's core 21700 NMC cylindrical cells and LFP chemistries, proving clear strategic alignment to supply these domains from its Hyderabad-based development facilities.

## InsideFPV Ventures Private Limited insideFPV ● Challengers X 80 Y 97 Overall 88

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 93/100

> InsideFPV shows clear self-reported capability and active development through its product listings and academic partnerships, but the evidence remains largely self-reported rather than independently verified.

SUPPORTING POINTS

- InsideFPV's product roadmap indicates a partnership with IITs aimed at pushing drone battery energy density up to 300 Wh/kg.
- The company provides an 8000mAh 6S 2P Lithium Ion battery pack that specifies a burst discharge rate of 20C.

#### Cycle Life & Durability — scored 73/100

> InsideFPV offers branded drone batteries but provides only vague, self-reported durability assertions rather than quantified cycle life ratings or third-party verification, leaving the evidence sparse. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- InsideFPV's commercial battery descriptions, such as for the Elevate V1 battery, list generic descriptions like "long service life" without providing specific cycle life counts, shelf life metrics, or environmental test results.
- Based on primary research, the company has not disclosed evidence regarding third-party verification, precise environmental testing data, or quantified cycle life ratings for InsideFPV's proprietary power solutions.

#### Safety & Thermal Management — scored 88/100

> The company shows clear capability in structural cell containment through a patented detachable frame and active thermal insulation for harsh environments, but lacks publicly verifiable evidence of formal third-party safety certifications like UN38.3 or IEC 62133 for its battery packs. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- InsideFPV Ventures Private Limited has been granted a patent for its "Detachable Battery Frame for Drone with Cover (Set)" by the Patent Office, Government of India, reflecting explicit R&D focus on structural integration and physical containment of the power unit.
- The company incorporates advanced battery insulation and thermal management features in its high-altitude surveillance systems (e.g., the Elevate Z1) to ensure safe operational resilience in extreme sub-zero conditions down to -20°C.

#### Customization & Form Factor Flexibility — scored 65/100

> The company primarily markets ready-to-fly (RTF) FPV and defense drones alongside retailing predefined battery models, lacking verified evidence of deep custom design services or form factor flexibility for energy storage units. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- InsideFPV provides standard off-the-shelf drone batteries under their own brand or third parties, such as the InsideFPV Lithium-Ion 2800mAh (4S/6S) and 8000mAh 6S 2P packs, rather than offering a dedicated custom battery design service.
- Based on primary research, the company has not disclosed evidence indicating that InsideFPV Ventures Private Limited provides specialized custom design services, variable form factor prototyping, or modular power management systems tailored specifically to external clients' mission profiles.

#### R&D & Innovation in Battery Chemistry — scored 82/100

> While InsideFPV has successfully patented mechanical drone components such as a detachable battery frame, based on primary research, the company has not disclosed proprietary innovation, research, or partnerships focused on advanced battery chemistry.

SUPPORTING POINTS

- InsideFPV Ventures Private Limited was granted a patent by the Government of India for a "Detachable Battery Frame for Drone with Cover (Set)".
- The company commercialises standard lithium-ion and lithium-polymer drone battery packs (such as 2800mAh 4S/6S and 8000mAh 6S formulations). Based on primary research, the company has not disclosed further details.

## Y — BUSINESS CAPABILITY

### OEM Partnerships & Customer Contracts — scored 93/100

> Secondary research confirms the company's long-term international MoU and a major short-term defense contract for fully integrated drone systems. Based on primary research, the company has not disclosed any explicit standalone battery supply agreements or OEM partnerships specifically restricted to its power storage units.

SUPPORTING POINTS

- InsideFPV successfully signed a Memorandum of Understanding (MoU) with Israel-based drone company UAV Dynamix to collaborate, exchange hardware systems, and co-explore the Indian and Israeli drone markets.

### Production Capacity & Scalability — scored 96/100

> Secondary research confirms the company's recent physical facility expansion to 20,000 sq. ft. and its proven capability to scale drone production rapidly for military contracts, but based on primary research, the company has not disclosed specific standalone metrics for battery manufacturing output, such as MWh/year,.

SUPPORTING POINTS

- InsideFPV expanded its manufacturing operations by moving to a new 20,000 sq. ft. facility to scale up production and support increased delivery demands.
- The company successfully scaled its manufacturing output to fulfill and execute a ₹10 crore Ministry of Defence contract within a strict two-month timeline, delivering hundreds of FPV/kamikaze drones to the Indian Army.

### Revenue Growth & Financial Health — scored 97/100

> Secondary research confirms the company's annual sales revenue, strong year-over-year growth rate, and recent venture capital funding status, but based on primary research, the company has not disclosed exact net profitability figures, so the financial disclosure falls short of full market-leading transparency.

SUPPORTING POINTS

- According to an Inc42 Company Profile, insideFPV Ventures Private Limited reported a sales revenue of ₹2.7 Cr in FY25, marking an 80.5% growth rate from the ₹1.5 Cr recorded in FY24.
- As reported by The Economic Times, the company raised ₹6 crore in a Pre-Series A funding round led by Gujarat Venture Finance Limited (GVFL) in January 2026 to fuel advanced hardware capability building and core platform product development.

### Geographic Reach & Distribution Network — scored 100/100

> Secondary research confirms the company's regional Indian presence via its offices and drone arenas, alongside its international reach through an export-focused cross-border partnership, but comprehensive independent distribution logs or exact service center counts specifically mapping localized drone battery supply agreements remain limited.

SUPPORTING POINTS

- InsideFPV maintains active operations across Indian regions by establishing over 30 drone experience arenas throughout the country and extending its localized infrastructure beyond its Surat, Gujarat corporate headquarters with an additional regional office in Gurgaon, Haryana.

### Certifications & Compliance for Defense/Aerospace — scored 100/100

> Secondary research confirms the company's operational execution under Indian defense procurement frameworks and active integration of aerospace compliance standards such as AS9100D, MIL-STD, and JSS; however, third-party certificate registries specifically isolated to its individual standalone battery packs, as opposed to complete UAV systems, remain self-reported or combined under total system deployment.

SUPPORTING POINTS

- InsideFPV operates within defense testing standards, detailing compliance and hiring technical expertise for aerospace and defense standards including AS9100D, JSS 0251, JSG 0852, and MIL-STD testing protocols for their deep-tech systems in Surat.
- The company successfully executed a ₹10 crore defense contract supplying combat/kamikaze drones to the Indian Ministry of Defence, demonstrating compliance with Indian defense procurement norms and DGCA Type Certification consultancy workflows.

## Aeroknots ● Challengers

X 84 Y 86 Overall 85

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 93/100

> While Aeroknots verifiably supplies high-discharge lithium-ion batteries and integrates real-time battery voltage monitoring systems for tactical and training UAV platforms, specific metrics on energy density (Wh/kg), distinct continuous power ratings, or numerical C-rates are entirely absent from public records, keeping this a partial capability.

SUPPORTING POINTS

- Aeroknots secured a government tender from the Government e-Marketplace (GeM) for a 10-inch Kamikaze drone integrated with a "High Discharge Lithium-Ion Battery," confirming power output capabilities tailored for aggressive flight profiles.
- A technical specification sheet for the AeroKnots 10-inch FPV drone details built-in power management safety integration, including real-time battery monitoring, system health checks, and automatic Return-to-Home (RTH) protocols triggered by low battery thresholds.

#### Cycle Life & Durability — scored 85/100

> Secondary research confirms the battery integration and high-altitude, rugged terrain envelope of the integrated systems via official contract filings and specification manuals, but based on primary research, the company has not disclosed third-party lab verification or data for specific cycle life ratings, shelf life, or humidity metrics, so standalone battery component performance remains unproven.

SUPPORTING POINTS

- An official product spec sheet for the AeroKnots 10-inch FPV drone details a ruggedized operational envelope engineered for high-altitude tactical operations up to 14,000 feet AMSL and mountain terrain stability, though it omits numerical battery cycle ratings Scribd.
- A government contract award via the GeM portal (GEM/2026/B/7187765) records that the Indian military procured custom high-discharge lithium-ion battery packs alongside a 10-inch Kamikaze UAV setup from AeroKnots, providing proof of integration into field-ready configurations Tenderkart.

#### Safety & Thermal Management — scored 67/100

> While basic operational safety redundancies like low-battery return-to-home were verified, public evidence documenting specialized battery thermal runaway mitigation or specific industry safety certifications for their custom assemblies is absent, leaving the firm toward the lower end of the scale.

SUPPORTING POINTS

- AeroKnots incorporates standard drone flight safety mechanisms, specifically an automated return-to-home protocol triggered upon communication loss or critical low-battery conditions, as detailed in the official AeroKnots 10-Inch FPV Tactical Surveillance UAV Specification Sheet.
- The company integrates basic lithium-based drone battery configurations to supply power for tactical operations like its 11-unit FPV surveillance deployment delivered to the 40th BN ITBP in Chhattisgarh, according to documentation on the official AeroKnots Platform.

#### Customization & Form Factor Flexibility — scored 94/100

> Secondary research confirms AeroKnots' capacity to supply custom high-discharge battery solutions for tactical drones and offer internal prototyping, but based on primary research, the company has not disclosed standardized modular metrics or third-party performance validations.

SUPPORTING POINTS

- AeroKnots advertises integrated custom design, rapid prototyping, and manufacturing services for drone power systems and UAV assemblies on its official corporate portal and commercial trade platforms.

#### R&D & Innovation in Battery Chemistry — scored 82/100

> The rating rests on the supporting points listed for this parameter. Based on primary research, the company has not disclosed further independently verified details.

SUPPORTING POINTS

- Not available

### Y — BUSINESS CAPABILITY

#### OEM Partnerships & Customer Contracts — scored 90/100

> Secondary research confirms two public sector/defense customer contracts involving drone batteries and related cases, but based on primary research, the company has not disclosed any long-term OEM supply agreements or named commercial drone manufacturer partnerships.

SUPPORTING POINTS

- AeroKnots was awarded a government tender on the Government e-Marketplace (GeM) in February 2026 valued at ₹1,12,000 for a 10-inch Kamikaze Drone, high-discharge Lithium-Ion batteries, and related components.
- AeroKnots secured an additional procurement contract from the Indian Department of Military Affairs in June 2026 valued at ₹49,000 to supply a UAV Battery Case, propellers, and specialized UAV components.

### Production Capacity & Scalability — scored 85/100

> While the presence of a localized assembly and testing facility in Lucknow was verified, specific manufacturing output in MWh/year and verifiable scaling capabilities could not be tracked, relying instead on partial, self-reported foundational setups. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- AeroKnots operates from an industrial facility located at Mansarovar Yojna, Kanpur Road, Sector O, Amausi, Lucknow, where they establish drone laboratory infrastructure, including drone assembly stations and indoor testing areas, to manufacture tactical, training, and custom systems.
- Based on primary research, the company has not disclosed information for this parameter.

### Revenue Growth & Financial Health — scored 78/100

> Secondary research confirms the baseline corporate legal scale and recent trajectory indicating a revenue decline, but based on primary research, the company has not disclosed specific product-level sales, external venture capital funding, or dedicated government grant figures for its custom battery assemblies.

SUPPORTING POINTS

- Not available

### Geographic Reach & Distribution Network — scored 81/100

> Secondary research confirmed the company's localized supply capabilities through domestic government tenders and its single physical office presence, but based on primary research, the company has not disclosed an extensive dealer, distributor, or export network, leaving it with a weaker standing here.

SUPPORTING POINTS

- A GeM contract award (Tender ID: GEM/2026/B/7187765) in February 2026 confirmed that Aeroknots secured a tender to supply high-discharge lithium-ion batteries and UAV systems within India.
- Public corporate registries document Aeroknots as operating out of a singular main facility situated in Mansarovar Yojna, Kanpur Road, Lucknow, Uttar Pradesh.

### Certifications & Compliance for Defense/Aerospace — scored 94/100

> Secondary research confirms AeroKnots' active compliance with Indian defense procurement norms via a verified GeM tender contract for battery-powered tactical drones, but independent third-party certificate listings for formal military standards (e.g., MIL-STD) or DGCA type-certifications specifically for its standalone battery assemblies could not be publicly cross-referenced. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Government e-Marketplace (GeM) tender award GEM/2026/B/7187765, where AeroKnots successfully secured a defense procurement contract to supply a 10-inch Kamikaze drone integrated with high-discharge lithium-ion batteries and related components.
- AeroKnots' official technical data sheet for its 10-inch FPV system, which outlines defense-grade operational profiles including a low-battery automatic Return-to-Home (RTH) failsafe mechanism and custom engineering for high-altitude military environments up to 14,000 feet AMSL.

## IPower Batteries Private Limited Ipower ● Challengers X 78 Y 90 Overall 84

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 67/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

#### Cycle Life & Durability — scored 71/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

#### Safety & Thermal Management — scored 85/100

> The company has partial, self-reported thermal management features and compliance frameworks across its broader lithium-ion catalog, but lacks distinct third-party safety verification specific to its drone energy products.

SUPPORTING POINTS

- IPower Batteries Private Limited manufactures lithium-ion packs and highlights advanced Battery Management Systems (BMS) with overcurrent protection, temperature control, and thermal pads between cells to dissipate heat and mitigate thermal runaway.
- Based on primary research, the company has not disclosed While the company holds certifications like IS 16046 (Part 2) / IEC 62133-2 for its light electric vehicle and storage packs, specific third-party safety or thermal certifications officially validated for its drone/UAV lineup.

#### Customization & Form Factor Flexibility — scored 89/100

> The company explicitly cites drone capabilities and a customized design ethos in its marketing collateral, but specific third-party verified evidence or case studies detailing custom drone battery prototyping, modularity, or dimensions is absent from public records.

SUPPORTING POINTS

- IPower Batteries Private Limited includes "Drone" in its official catalog listing of industries served, alongside claims of providing customization and a "tailormade" approach to meeting specific voltage and capacity requirements.

#### R&D & Innovation in Battery Chemistry — scored 77/100

> The company has general battery R&D capabilities for terrestrial automotive and storage markets, but lacks public evidence of targeted chemistry innovations for drone applications, a weak point on this measure. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- IPower Batteries Private Limited operates a government-recognized in-house R&D center in Kundli, Haryana, though its applications focus primarily on electric vehicles, energy storage systems, and lead-acid or lithium-ion pack designs rather than UAV-specific chemistries.
- Based on primary research, the company has not disclosed evidence regarding specialized patents, next-generation chemistry R&D partnerships, or advanced formulations like solid-state or lithium-sulfur explicitly engineered for unmanned aerial vehicles.

### Y — BUSINESS CAPABILITY

#### OEM Partnerships & Customer Contracts — scored 82/100

> While the company has verified, active OEM partnerships and customer testimonials within the broader clean-energy electric mobility and smart vehicle ecosystem, no long-term supply agreements, specific contract values, or client listings could be verified explicitly for drone manufacturers (UAVs) or defense organizations.

SUPPORTING POINTS

- The company signed a strategic alliance with e-mobility franchise network Electric One to supply customized IoT-enabled lithium-ion batteries across its retail chains and multi-brand EV ecosystem, according to The Times of India Press Release.
- The company has verified OEM partnerships and continuous collaborative supply agreements for electric mobility applications with regional vehicle manufacturers, specifically named clients Lohia Industries Ltd and Deltic, according to the IPower Batteries Official Website.

### Production Capacity & Scalability — scored 96/100

> Secondary research confirms the overall physical footprint, automated production lines, and factory capacity metrics from company data, but based on primary research, the company has not disclosed independent third-party audits dedicated specifically to UAV energy units, leaving this a self-reported capability.

SUPPORTING POINTS

- IPower Batteries Private Limited operates a 50,000+ square foot IMS-certified manufacturing facility in Kundli, Sonipat, Haryana, featuring an annual production capacity of 1.2 GWh as stated in the company's official corporate catalog.
- The company utilizes automated production lines with a self-reported active output of over 500 to 1,000 battery packs per day, supported by an in-house government-approved R&D center and approximately 300 employees.

### Revenue Growth & Financial Health — scored 86/100

> Secondary research confirms the overall sales revenue, profitability metrics, and funding status through third-party aggregator filings, but could not isolate drone-specific revenue segments, since the financials are aggregated under general electric vehicle and lightweight mobility applications; negative growth rates leave drone-specific financial performance unproven. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Official registry and corporate records compiled by Tracxn and Tofler show that IPower Batteries Private Limited generated an operating revenue range between ₹50 Cr and ₹100 Cr for the financial year ending March 31, 2025, but experienced a year-over-year revenue contraction of 62.38% and a net profit decline of 70.84%.
- According to Tracxn's platform records, the company remains classified as an entirely unfunded corporate entity that has not raised venture capital or institutional investment rounds, relying instead on its bootstrapped corporate operations and a paid-up capital base of ₹1 Cr.

### Geographic Reach & Distribution Network — scored 92/100

> Secondary research confirms the corporate footprint of the manufacturing facility and national sales/service network, but drone-specific regional distribution or third-party verified international distribution frameworks for the UAV market could not be confirmed through secondary research, leaving global claims self-reported. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- IPower Batteries operates a single 50,000 sq. ft. automated manufacturing facility in Kundli, Sonipat, Haryana, and relies on an internal network of 200+ sales and service centers across India to support its green energy products (About - Ipower Batteries, New Front Page).
- Based on primary research, the company has not disclosed while the company self-reports a global footprint spanning 50 countries with products reaching over 100 countries (Export - Ipower Batteries), specific distribution agreements, regional drone partnerships, or dedicated service networks explicitly designated for its "DRONE" battery catalog line.

### Certifications & Compliance for Defense/Aerospace — scored 94/100

> Secondary research confirms the company's core ISO management credentials (ISO 9001, 14001, 45001) and explicit technical testing capabilities aligned with MIL-STD-810 and DGCA/DGQA frameworks, but based on primary research, the company has not disclosed a finalized formal type certification listing specifically for drone battery models or distinct defense procurement contract fulfillment.

SUPPORTING POINTS

- A published professional installation report indicates that iPower Batteries provides drone and robot testing frameworks designed to meet Director General of Quality Assurance (DGQA), Directorate General of Civil Aviation (DGCA), Quality Council of India (QCI), and MIL-STD-810/JSS-55555 standard guidelines.
- Corporate registration documents and technical presentations from their Kundli facility verify that the plant formally holds Quality Management System certification under ISO 9001:2015, Environmental Management System certification under ISO 14001:2015, and Occupational Health and Safety certification under ISO 45001:2018.

## Aeroarc Private Limited Aeroarc ● Challengers X 74 Y 93 Overall 84

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 74/100

> The company plans to establish its own battery production facility to resolve industry supply bottlenecks, but specific technical documentation, data sheets, third-party verification, or explicit specs on energy density and power output metrics are missing from public documentation.

SUPPORTING POINTS

- In an interview regarding strategic partnerships, Aeroarc's MD stated that the company is setting up its own battery manufacturing unit in India to tackle the issue of battery shortage faced by the drone industry, but based on primary research, the company has not disclosed granular performance metrics such as Wh/kg or C-rates.

#### Cycle Life & Durability — scored 65/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

#### Safety & Thermal Management — scored 65/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Aeroarc Private Limited manufactures bespoke battery solutions for niche UAVs, including a modular battery for the Trinetra drone.

#### Customization & Form Factor Flexibility — scored 72/100

> The company has announced intent to set up its own battery manufacturing facility to solve supply constraints. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Aeroarc's Managing Director, Arjun Aggarwal, stated that the company is setting up its own battery manufacturing unit in India to address battery shortages and ramp up drone production.
- Based on primary research, the company has not disclosed evidence demonstrating specific custom design services, modular power management options, or rapid prototyping capabilities for drone batteries.

#### R&D & Innovation in Battery Chemistry — scored 92/100

> The company shows verifiable capability in supplying tailored drone batteries and adaptive power management software for its own systems, but lacks public evidence of primary R&D or proprietary development in next-generation chemistries. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Aeroarc Private Limited manufactures modular batteries for its UAV systems, such as the Modular Battery for Trinetra Drone and RPVA Trinetra Drone Battery, which have been qualified for Indian government procurement bids via the GeM portal.
- In its technical presentations, the company features power management capabilities that allow adjusting parameters to accommodate new or existing battery chemistries to optimize life and safety. Based on primary research, the company has not disclosed specific advanced chemical formulations.

### Y — BUSINESS CAPABILITY

#### OEM Partnerships & Customer Contracts — scored 94/100

> The company has verified prime customer contracts with defense organizations like the Indian Army and manufacturing partnerships with global OEMs, but these agreements focus on localized drone platform manufacturing and co-development rather than explicit, standalone third-party supply contracts for its battery technologies.

SUPPORTING POINTS

- Aeroarc established a major international OEM manufacturing partnership with U.S. drone manufacturer Skydio to locally manufacture and support secure, autonomous small uncrewed aircraft systems (sUAS) for Indian government customers.

### Production Capacity & Scalability — scored 94/100

> The company's dual facilities, global supply partnerships, and significant expansion capital were verified. Based on primary research, the company has not disclosed its specific battery factory size, dedicated cell production lines, or MWh/year output for energy storage units.

SUPPORTING POINTS

- Aeroarc Private Limited's LinkedIn Profile confirms that the company maintains over 100 professionals distributed across two state-of-the-art manufacturing facilities within India to develop its aerial and terrestrial unmanned systems.
- Public corporate data and global partnership records show that Aeroarc has secured an initial ₹40 crore strategic investment from Japanese drone major ACSL to bolster its local robotics supply chain and target $30 million in UAV projects, while partnering with US-based Ghost Robotics for localized advanced manufacturing.

### Revenue Growth & Financial Health — scored 86/100

> Third-party records verified absolute sales revenue and funding status, but independent annual profitability data and targeted government grants for battery operations remained unverified. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Inc42 Profile Data confirms that Aeroarc Private Limited reported a steep decline in revenue to ₹8.0 Cr ($951K) for FY25 down 96.4% from ₹223.6 Cr in FY24.
- Tracxn Financial Records report that the company raised a total of $4.79 million in funding, highlighted by investments to strengthen drone and robotics supply lines.

### Geographic Reach & Distribution Network — scored 97/100

> International trade channels and a major joint manufacturing distribution agreement for the Indo-Pacific region were verified, but based on primary research, the company has not disclosed granular public evidence for localized domestic distributors, dealers, or service centers.

SUPPORTING POINTS

- Eximpedia's Import-Export Trade Data demonstrates that Aeroarc Private Limited actively participates in international trade under HS Code 88062200, with established import-export channels predominantly linking to the United States and Hong Kong via the Delhi Air Cargo gateway.
- The Times of India details a strategic global manufacturing and distribution partnership between Aeroarc and US-based drone manufacturer Skydio, targeting defense customers across India and the wider Indo-Pacific region.

### Certifications & Compliance for Defense/Aerospace — scored 94/100

> Secondary research confirms the company's official DGCA platform registration and general ISO 9001 status, but found only broad self-reported claims regarding battery-specific compliance standards.

SUPPORTING POINTS

- The Directorate General of Civil Aviation (DGCA) inventory of approved Remotely Piloted Aircraft Systems (RPAS) lists Aeroarc Private Limited as an active manufacturer for its small category rotorcraft model, "Aero krishi / AK-HEX-8L".
- B2B industrial market listings for Aeroarc's drone systems indicate a corporate compliance standard of ISO 9001 for quality management systems.

## Pastiche Energy Solutions Private Limited

Pastiche · Challengers · X 75 · Y 91 · Overall 83

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 84/100

> The company mentions providing custom lithium drone batteries with high energy density, but all available product literature lacks any specific, verifiable data on Wh/kg metrics, continuous power output, or C-rate performance. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Pastiche Energy Solutions self-reports manufacturing a "premium quality range of Lithium Battery For Drones" with generic attributes of being "Rechargeable," having a "Long life," and featuring "High energy density," but does not list specific numerical specifications such as Wh/kg or C-rate.

#### Cycle Life & Durability — scored 65/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. Based on primary research, the company has not disclosed drone batteries produced by Pastiche Energy Solutions Private Limited.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

#### Safety & Thermal Management — scored 65/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. Based on primary research, the company has not disclosed safety certifications, such as UN38.3 or IEC 62133, or specific thermal runaway mitigation designs for Pastiche Energy Solutions' drone-specific batteries.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

#### Customization & Form Factor Flexibility — scored 92/100

> The company provides self-reported evidence of custom configuration capabilities for its battery packs, but it lacks third-party verified credentials or detailed documentation on rapid prototyping or modular configurations optimized for diverse drone mission profiles. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Pastiche Energy Solutions explicitly highlights on its official website that their battery systems are "Customizable" with "Tailored solutions to meet your specific energy storage requirements".
- According to primary research, third-party B2B marketplace profiles list Pastiche Energy Solutions as a provider of "Customized battery packs" alongside a generic offering of "Lithium Battery For Drones".

#### R&D & Innovation in Battery Chemistry — scored 68/100

> The company lacks verifiable third-party or self-reported evidence of advanced R&D investment or proprietary innovation in battery chemistry specific to drones. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed evidence indicating custom chemical research or patents held by the company for advanced UAV-specific battery chemistries.
- Pastiche Energy Solutions operates as an assembler and packer utilizing standard cylindrical, prismatic, and lithium polymer cells sourced from third-party manufacturers.

### Y — BUSINESS CAPABILITY

#### OEM Partnerships & Customer Contracts — scored 83/100

> Generalized vendor status and equity partnerships are documented, but the absence of publicly verifiable long-term supply agreements or named drone manufacturing clients limits the company's standing. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Pastiche Energy Solutions acts as a registered vendor supplying custom energy units to select Indian government and private organizations across the Aerospace, Defense, and Marine sectors, as detailed on the official company portal.
- The company maintains structural equity ties with Bhago Mobility Solutions, accessing industrial networks linked to Japanese and global automotive OEMs.

### Production Capacity & Scalability — scored 100/100

> Secondary research confirms the operational status of the Panchkula assembly facility and the Punjab plant expansion plans, but based on primary research, the company has not disclosed exact annual output metrics specifically for the drone battery division, since the underlying data is self-reported.

SUPPORTING POINTS

- Pastiche Energy Solutions operates an assembly and packaging facility at Plot No. 46, Industrial Area, Phase 2, Panchkula, Haryana, featuring production and testing lines for high-discharge lithium polymer packs.

### Revenue Growth & Financial Health — scored 97/100

> Secondary research confirms the company's precise annual sales revenue, annual growth rate, and strategic funding status through official registry databases and press tracking. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- According to corporate filings tracked by The Company Check, Pastiche Energy Solutions Private Limited reported annual sales revenue of ₹3.58 Cr for FY 2024, achieving a year-over-year revenue growth rate of 59.00%.

### Geographic Reach & Distribution Network — scored 92/100

> The company's dual northern Indian operations facilities in Haryana and Punjab were successfully verified. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- According to primary research, the company self-reports a global supply aspiration and domestic "wide distribution network" paired with strategic partnerships for EV infrastructure, tracked across wholesale listing spaces including its corporate profile.

### Certifications & Compliance for Defense/Aerospace — scored 83/100

> Self-reported quality certifications and defense-targeted hardware listings were verified, but based on primary research, the company has not disclosed active third-party aerospace or defense-specific credentials such as DGCA approvals or MIL-STD compliance.

SUPPORTING POINTS

- Pastiche Energy Solutions Private Limited holds an ISO 9001:2008 quality certification, which is self-listed across its standard industrial corporate documentation and early distributor frameworks.
- The company markets specialized defense lithium battery packs and aerospace power systems directly to military and regional equipment supply networks.

## AKSI AI Aerospace and Agri Technology Ventures Private Limited LiHi ● Trailblazers X 91 Y 83 Overall 87

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 95/100

> Secondary research confirms the company's module capacities and supported flight endurance ratings, but based on primary research, the company has not disclosed independent third-party lab certifications for exact energy density (Wh/kg) or peak continuous discharge C-rates, leaving these metrics self-reported.

SUPPORTING POINTS

- LiHi Smart Battery Technologies Private Limited (a subsidiary of AKSI Aerospace Group) produces modular configurations including a 6S 25,000mAh smart battery pack designed for the Skywing 90 drone platform (Source: Aksi Aerospace corporate documentation).
- The company's product lineup features high-capacity 30Ah and 50Ah smart battery packs engineered to deliver continuous flight endurance ranging from 70 to over 90 minutes for industrial and agricultural UAV applications (Source: Aksi Aerospace product overview).

#### Cycle Life & Durability — scored 94/100

> Secondary research confirms the integration of protective management software and a field-durable physical chassis layout to shield against extreme conditions, but based on primary research, the company has not disclosed specific empirical data on exact cycle counts, shelf life parameters, or certified laboratory temperature/humidity thresholds.

SUPPORTING POINTS

- AKSI Aerospace Group's proprietary LiHi Smart Batteries feature integrated overcharge and over-discharge protection systems built into the power management setup specifically designed to preserve and extend the cell's functional life cycles.
- AKSI Aerospace designs its LiHi battery line with a ruggedized and field-durable housing intended to protect the internal energy storage modules against physical stress and extreme external environmental field conditions.

#### Safety & Thermal Management — scored 84/100

> While the group shows public active involvement in manufacturing indigenous drone batteries, specific third-party safety certifications (e.g., UN38.3, IEC 62133) or documented thermal runaway test data could not be confirmed through secondary research online, leaving its compliance self-reported. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Exposure at the AEDEX + AIRPORT Expo 2026, where the group showcased its capabilities across drones, flight control systems, and indigenously developed battery systems for defense and commercial sectors.
- Corporate records indicate the presence of a dedicated sister subsidiary under the same management and address, LIHI Smart Battery Technologies Private Limited, registered to drive the group's custom power pack engineering.

#### Customization & Form Factor Flexibility — scored 92/100

> Secondary research confirms the company's ability to offer capacity tiers optimized for distinct mission types and modular compatibility across different aerial configurations, but based on primary research, the company has not disclosed third-party evidence of custom-built form factors, bespoke dimension tailoring, or rapid prototyping services.

SUPPORTING POINTS

- The company's specialized battery division, LiHi Smart Batteries, offers an optimized product lineup consisting of specific capacities (20Ah, 30Ah, and 50Ah) tailored to varying mission profiles like lightweight flights or extended long-range cargo operations, as detailed on the LiHi Smart Batteries - Aksi Aerospace.
- LiHi Smart Batteries feature structural plug-and-play compatibility designed to interface natively with multiple distinct UAV form factors, including multi-rotor, fixed-wing, and VTOL platforms.

#### R&D & Innovation in Battery Chemistry — scored 92/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- LiHi Smart Batteries (a subsidiary of AKSI Aerospace Group) showcased its indigenously developed lithium-ion battery configurations (ranging from 20Ah to 50Ah packs) at the 6th International Drone Expo in New Delhi, demonstrating a customized Battery Management System (BMS) optimized for high thermal safety and cell balancing in extreme climates like Ladakh and Rajasthan.

## Y — BUSINESS CAPABILITY

### OEM Partnerships & Customer Contracts — scored 100/100

> Secondary research confirms the major global OEM contract with FIXAR encompassing drone batteries as well as defense-related deployments. Based on primary research, the company has not disclosed specific financial allocations strictly isolating battery packs from overarching subsystems.

SUPPORTING POINTS

- AKSI Aerospace Group signed a strategic agreement valued at ₹85 crore with Latvia-based global unmanned aerial systems manufacturer FIXAR Global at Aero India 2025 to supply critical indigenously developed subsystems, including drone batteries.
- AKSI Aerospace Group reported that its drone components, including its specialized LiHi Smart Batteries division, were utilized in field operations by the Indian Armed Forces.

### Production Capacity & Scalability — scored 65/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. Based on primary research, the company has not disclosed its manufacturing output, facility metrics, and scaling capabilities, given a complete lack of public documentation or verifiable operational data.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed information for this parameter.

### Revenue Growth & Financial Health — scored 91/100

> Secondary research confirms the company's forward-looking order pipeline, projected revenue targets, and facility investment plans through commercial announcements, but based on primary research, the company has not disclosed exact historical audited profit figures or detailed official annual report breakdowns specifically isolating the standalone battery entity's profitability.

SUPPORTING POINTS

- AKSI Aerospace Group secured a ₹85 crore ($10 million) contract from Latvia-based FIXAR Global to supply drone subsystems—including its LiHi smart batteries, battery management systems (BMS), and autopilots—and announced a planned $5 million investment to expand its Hyderabad facility.
- Managing Director Pankaj Akula stated that the firm anticipated a twofold increase in revenue to ₹70 crore for the next fiscal year, up from an estimated ₹35 crore in the concurrent fiscal period.

### Geographic Reach & Distribution Network — scored 76/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Official registrar and corporate tracking data from Zauba Corp and Instafinancials verifies the company's registered head office at Secunderabad, Hyderabad, Telangana, India.
- Exhibition records from the AEDEX + AIRPORT Expo 2026 Exhibitor List confirm that the company officially showcased its drone batteries and power solutions at booth B-534 in New Delhi.

### Certifications & Compliance for Defense/Aerospace — scored 82/100

> Secondary research confirms the company's stated focus on meeting DGCA compliance frameworks for its smart batteries, but based on primary research, the company has not disclosed third-party evidence or specific certificate registration numbers for MIL-STD or ISO 9001 compliance, leaving formal defense certifications unproven.

SUPPORTING POINTS

- The company designs and manufactures its LiHi Smart Batteries and subsystem configurations to comply with emerging DGCA standards, targeting operational excellence and field-proven safety systems as detailed by its subsidiary Dronevation (Source: AKSI Aerospace Group).
- While AKSI Aerospace Group has established a 50,000 to 100,000 sq. ft. vertically integrated manufacturing facility in Hyderabad and handles defense-aligned subsystems, public product filings and documentation for the battery division do not list formal, active MIL-STD or ISO 9001 certifications (Source: AKSI Aerospace Group).

## QuillAir Aerospace and Engineering Private Limited QuillAir Trailblazers X 90 Y 82 Overall 86

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 95/100

> Secondary research confirms the company's self-reported battery capacity (501.6Wh) and continuous discharge rate (25C), but based on primary research, the company has not disclosed independent, third-party laboratory evaluations of the actual energy density (Wh/kg) or exact weight metrics, leaving this a self-reported capability.

SUPPORTING POINTS

- According to primary research, the company offers the QuillAir EP T 22000mAh 6S Li-Po battery designed for agricultural and industrial drones, which features a specified capacity of 501.6Wh and a discharge rate of 25C, according to its official retail profile.
- According to primary research, the company self-reports its technical specifications on its public vendor profiles, detailing a nominal voltage of 22.8V and a 6S configuration for its heavy-payload agricultural applications.

#### Cycle Life & Durability — scored 91/100

> Secondary research could only confirm self-reported marketing claims of a long cycle life and durability; based on primary research, the company has not disclosed specific numerical cycle life ratings, shelf life, or independent environmental test results.

SUPPORTING POINTS

- According to primary research, the company self-reports that its QuillAir EP T 22000mAh 6S 22.2V Li-Po Battery, designed for agricultural and industrial drones, offers a long cycle life and efficient performance in field conditions, as listed on its company profile.
- General documentation on QuillAir's Official Website states that its custom-manufactured unmanned systems are built with sustainable energy solutions and engineered for durability and reliability.

#### Safety & Thermal Management — scored 84/100

> The presence of integrated protection circuitry was verified on the main commercial drone battery pack, but third-party validation, international safety certifications such as UN38.3 or IEC 62133, and based on primary research, the company has not disclosed thermal runaway mitigation features, leaving this a partial, self-reported strength.

SUPPORTING POINTS

- According to primary research, QuillAir Aerospace and Engineering Private Limited states that its QuillAir EP T 22000 MAH Agri Drone Battery features a built-in Protection Circuit Board (PCB) for safety threshold governance, as listed on its business profile.
- The company reports that its agricultural and custom drone systems utilize integrated energy solutions designed for field endurance, according to its corporate profile.

#### Customization & Form Factor Flexibility — scored 94/100

> Secondary research confirms the availability of a dedicated high-capacity drone battery model and self-reported industry-specific customization services, but based on primary research, the company has not disclosed independent third-party evidence of rapid prototyping capabilities or custom design options for external OEMs.

SUPPORTING POINTS

- According to primary research, the company offers the QuillAir EP T 22000 MAH Agri Drone Battery, a dedicated 6S 22.2V Li-Po battery pack providing a 501.6Wh capacity and 25C discharge rate with a built-in protection PCB tailored for heavy agricultural and industrial missions, as detailed on its business profile.
- The firm claims industry-specific customization and provides advanced battery solutions alongside its custom drone manufacturing services to adapt platforms for agriculture, defense, and smart city sectors, according to its QuillAir.

#### R&D & Innovation in Battery Chemistry — scored 84/100

> Secondary research confirms that the company supplies dedicated high-capacity drone battery packs. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- According to primary research, the company commercialises the QuillAir EP T 22000 MAH Agri Drone Battery, which relies on standard Lithium Polymer (LiPo) cell chemistry with a 25C discharge rate and 501.6 Wh power output, as verified via its product catalog.

## Y — BUSINESS CAPABILITY

### OEM Partnerships & Customer Contracts — scored 71/100

> Based on primary research, the company has not disclosed any named OEM partnerships, defense organization clients, contract values, or long-term supply agreements for the company's battery units, as its available public information is limited to generalized commercial product listings and self-reported manufacturing descriptions.

SUPPORTING POINTS

- According to primary research, QuillAir Aerospace and Engineering Private Limited markets its proprietary "QuillAir EP T 22000 MAH Agri Drone Battery" directly via B2B marketplaces, indicating independent commercial manufacturing capabilities.
- Based on primary research, the company has not disclosed information for this parameter. No third-party verified, named long-term supply agreements or corporate/defense customer contracts for its battery packs are available in public disclosures or company channels.

### Production Capacity & Scalability — scored 86/100

> Based on primary research, the company has not disclosed its exact manufacturing throughput metrics, dedicated assembly lines, or official industrial expansion plans to support large-scale enterprise orders.

SUPPORTING POINTS

- According to primary research, QuillAir Aerospace and Engineering Private Limited markets individual commercial units of specialized drone batteries, specifically the "QuillAir EP T 22000 MAH Agri Drone Battery" (6S 22.8V, 501.6 Wh LiPo pack) via e-commerce channel distributions.
- Corporate registry documents and company filings outline that the entity was incorporated in March 2025 with a limited paid-up capital of ₹10.00 Lakh and an employee base estimated at 11–50 staff operating from its registered facility in Garden Reach, Kolkata.

### Revenue Growth & Financial Health — scored 78/100

> While basic incorporation capital and retail pricing are verified, annual sales revenue, growth rates, and based on primary research, the company has not disclosed formal funding, given the lack of public financial disclosures.

SUPPORTING POINTS

- QuillAir Aerospace and Engineering Private Limited has a registered corporate authorized share capital of ₹10 Lakhs and a paid-up capital of ₹10 Lakhs following its March 29, 2025 incorporation, as tracked by Tracxn.
- According to primary research, the company self-lists its custom QuillAir EP T 22,000 mAh Agri Drone Battery pack for retail at a unit price of ₹13,000 on a B2B marketplace.

### Geographic Reach & Distribution Network — scored 81/100

> Secondary research confirms the company's localized presence, centralized manufacturing facility in Kolkata, and active deployment of online B2B direct distribution channels; however, based on primary research, the company has not disclosed any physical regional branch locations, dedicated physical dealerships, or international export market footprints, limiting its evaluation to localized operations with self-managed distribution.

SUPPORTING POINTS

- A manufacturing and operations facility established at Building No.-23, 43/1 Garden Reach Road, Rajabagan Dock Yard, Kolkata, West Bengal, India, which serves as the central hub for their engineering, assembly, and customer services.
- According to primary research, commercial retail listings and direct sales point validation via a B2B digital marketplace, through which the company distributes its proprietary "QuillAir EP T 22000 mAh 6S Agri Drone Battery" directly from its West Bengal base to nationwide corporate or agricultural buyers.

### Certifications & Compliance for Defense/Aerospace — scored 93/100

> Secondary research confirms that the company has actively undertaken the compliance, documentation, and external coordination processes for aerospace standards like AS9100 and ISO 9001, though final third-party certificate listings for the batteries themselves remain partial or self-reported rather than fully published.

SUPPORTING POINTS

- The company has documented internal organizational preparation and coordination for key industry standards including ISO 9001, AS9100, BIS, and DIL documentation activities via external certification agencies.
- The company actively participates in and tracks bids for the Government e-Marketplace procurement frameworks, and enforces mandatory DGCA certification rules internally for its operational workflows.

## Techlanz Private Limited Vouwl ● Trailblazers X 87 Y 86 Overall 86

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 65/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

#### Cycle Life & Durability — scored 94/100

> Techlanz presents concrete performance ratings and specific product specifications for its drone batteries, but the evidence is primarily self-reported and lacks publicly shared third-party certification. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Techlanz Private Limited markets its smart drone batteries under the VOUWL brand, claiming specific cycle-life enhancements such as "40% More Cycle Life" and showcasing models like the Boltz-6S 22,000mAh battery rated for "Up to 800 Charge Cycles".
- The company publicises that its drone technology delivers a high cycle life protected by a rugged hard casing or lightweight aluminum enclosure, though detailed third-party environmental test documentation remains private.

#### Safety & Thermal Management — scored 85/100

> Techlanz has clear internal design capabilities for drone battery safety and thermal modeling. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Techlanz launched its purpose-built drone battery platform, "Vouwl", utilizing drone-specific Battery Management System (BMS) hardware and firmware algorithms to specifically address safety, predictability, and lifespan issues.
- The company actively designs battery thermal management systems and performs battery thermal modeling and simulations, displaying its capabilities at Drone Expo 2026.

#### Customization & Form Factor Flexibility — scored 94/100

> The company has verifiable capabilities in designing custom-grade battery architecture and specialized battery management systems for distinct drone operational types, though detailed public data on automated rapid prototyping and modular hardware ranges stays internal, consistent with its early-stage seed footprint.

SUPPORTING POINTS

- Techlanz Private Limited operates the drone battery platform "Vouwl," offering smart drone-specific battery hardware, firmware, and software stacks tailored to target sectors like agriculture, surveillance, and defence drones.
- Job openings for technical sales and business development at the company state responsibilities include translating specific customer statement of requirements (SORs) and requests for quotes (RFQs) into feasible solutions, providing market feedback to support product customization.

#### R&D & Innovation in Battery Chemistry — scored 98/100

> Techlanz shows clear, verifiable product development capabilities for drone power management and cell selection under its Vouwl platform. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Techlanz Private Limited launched a dedicated drone battery platform called Vouwl, which focuses on smart battery technology, custom cell selection, and proprietary cell-sorting algorithms to maximize performance in agricultural, defense, and surveillance UAVs.
- The company actively invests in next-generation drone battery development by hiring specialized BMS Firmware Engineers and Battery Pack Engineers focused on safety-critical embedded systems and advanced cell modeling (including RC and physics-based methodologies).

### Y — BUSINESS CAPABILITY

#### OEM Partnerships & Customer Contracts — scored 86/100

> Initial delivery capabilities to an unnamed OEM are noted, but the absence of named commercial clients, defense contracts, or long-term supply agreement valuations limits the company's standing. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Co-founder Vishal Ahuja confirmed that Techlanz Private Limited shipped its first production batch of "Vouwl" battery packs to an unnamed Drone OEM.
- Techlanz was selected as one of 10 startups in Karnataka under the Government of Karnataka and Government of France Global Innovation Program, requesting international business collaboration and partnerships.

### Production Capacity & Scalability — scored 92/100

> Secondary research confirms that Techlanz has transitioned into an active product manufacturer with an initial drone battery production run and local assembly technicians. Based on primary research, the company has not disclosed its numerical manufacturing output in MWh/year and explicit scaling plans.

SUPPORTING POINTS

- Techlanz Private Limited announced the commercial launch of its drone battery platform, "Vouwl," in mid-2026 and officially shipped its first production battery packs to a drone OEM in late April 2026.
- The company operates from a single facility located in Bangalore North, where it has actively posted job descriptions to hire local "Battery Production Technicians" for electrical module assembly and spot welding.

### Revenue Growth & Financial Health — scored 91/100

> Secondary research confirms the company's early-stage venture funding status and annual revenue range through corporate registry records and announcements, but based on primary research, the company has not disclosed profitability or detailed growth rates given its short operating history.

SUPPORTING POINTS

- Official regulatory filings registered with the Ministry of Corporate Affairs (MCA) via Tracxn show that Techlanz Private Limited generated under ₹10 Crore in annual revenue for the financial year ending on March 31, 2025.
- The company announced a capital injection of ₹1.5 Crore in Angel Funding supported by incubation under the TiHAN Startup Program at IIT Hyderabad.

### Geographic Reach & Distribution Network — scored 79/100

> Secondary research confirms that the company has begun initial commercial fulfillment to an OEM from its single operational facility in Bangalore. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Co-founder Vishal Ahuja publicly confirmed that Techlanz shipped its first commercial production batch of "Vouwl" drone battery packs directly from its headquarters to an unnamed Drone OEM.
- Techlanz operated and marketed its drone battery technologies locally out of a single primary corporate and R&D facility located at Srirampura, Jakkur in Bangalore, Karnataka.

### Certifications & Compliance for Defense/Aerospace — scored 83/100

> Secondary research confirms that the firm markets smart drone battery capabilities to the defense and surveillance sector under the Vouwl platform and hires engineers to work on local Indian automotive/electronics standards such as AIS 156 and BIS IS 17855. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Techlanz Private Limited operates its drone-focused division under the proprietary product brand Vouwl, which explicitly targets and positions its custom smart BMS hardware, firmware, and algorithms for applications in surveillance and defense drones.
- In its technical organizational structure and recruitment guidelines, Techlanz seeks engineers to manage battery verification protocols under automotive/EV standards like AIS 156, performance criteria such as BIS IS 17855, and standard transport validations including UN 38.3, demonstrating underlying alignment with foundational lithium validation rules.

## Lixir Energy Private Limited LIXIR ● Trailblazers X 88 Y 84 Overall 86

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 91/100

> Lixir Energy presents self-reported commercial capability, including general descriptions of high energy density and high discharge rates, but it lacks specific verified Wh/kg figures or independent third-party certification.

SUPPORTING POINTS

- Not available

#### Cycle Life & Durability — scored 91/100

> The company self-reports a general 1.7x extension in battery life from its proprietary thermal management. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Lixir Energy claims its patent-pending thermal management technology enhances overall battery life by 1.7 times.
- The company advertises lightweight, high-energy-density battery packs specifically for military drones, but does not publish specific cycle life ratings, environmental tolerances, or third-party durability test results for these drone units.

#### Safety & Thermal Management — scored 88/100

> The company shows present, award-recognized internal capability for thermal management, but the lack of publicly documented third-party safety certifications, such as UN38.3 or IEC 62133, limits its standing. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Lixir Energy features a patent-pending thermal management technology combined with a proprietary smart battery management system (BMS) to mitigate thermal runaway and extend battery life.
- The company received the "Best Innovative Product Award" at the 6th National Conference on Innovative Global Technology Trends in March 2024 for its battery thermal management technology and safety against thermal runaway.

#### Customization & Form Factor Flexibility — scored 76/100

> The company clearly manufactures drone batteries as part of its energy storage portfolio, but its public documentation lacks specific verification or explicit claims regarding custom engineering services, rapid prototyping, or geometric form factor flexibility for UAVs. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Lixir Energy Private Limited manufactures lightweight and high-energy-density battery packs for military drones alongside commercial LiPo drone batteries available through consumer platforms.
- Based on primary research, the company has not disclosed information for this parameter.

#### R&D & Innovation in Battery Chemistry — scored 92/100

> The company demonstrates internal product assembly capabilities featuring self-reported patent-pending thermal management hardware, but lacks public evidence of R&D investment or breakthroughs in fundamental advanced chemistries like solid-state or lithium-sulfur. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Lixir Energy Private Limited states on its official portal that it develops lightweight, high-energy-density drone battery packs utilizing a patent-pending thermal management technology that enhances battery life by 1.7 times.
- Commercial retail listings for its standard drone battery line show reliance on standard lithium polymer (LiPo) cells rather than next-generation battery chemistries.

### Y — BUSINESS CAPABILITY

#### OEM Partnerships & Customer Contracts — scored 82/100

> The company self-reports general client satisfaction and lists military drone batteries commercially, but it was not possible to verify any named drone manufacturers, defense clients, or long-term supply contracts.

SUPPORTING POINTS

- Lixir Energy Private Limited's website features general self-published quotes from individual professionals, such as "Priya Sharma, Aerospace Engineer," praising battery solutions without disclosing specific enterprise or organizational clients.
- According to primary research, commercial listings document that the company offers "Batteries for Military Drones" at a price point of ₹99,999/Piece, though no defense organization or supply contract is referenced.

### Production Capacity & Scalability — scored 86/100

> Secondary research confirms the company's general micro-scale facility indicators and small employee base. Based on primary research, the company has not disclosed any official MWh/year output metrics, distinct drone-specific production lines, or formalized scale-up plans tailored to large-scale military or industrial UAV orders, leaving this a weaker, self-reported area.

SUPPORTING POINTS

- According to operational records from ExportersIndia, Lixir Energy Private Limited manufactures custom inverter battery packs at its Pune facility with a listed production capacity of 100 batteries per day.
- According to primary research, supplier directory listings show that Lixir Energy operates as a small-scale manufacturer and service provider with a workforce of 21 to 50 employees and an estimated annual turnover of Rs. 0.5 to 2.5 Crore.

### Revenue Growth & Financial Health — scored 89/100

> The company's baseline revenue figures and growth rates were verified via corporate registries, but external venture capital rounds or government grants remain unconfirmed.

SUPPORTING POINTS

- Official regulatory filings analyzed by Tracxn confirm that Lixir Energy Private Limited generated less than ₹10 Cr in total revenue for the financial year ending March 31, 2025, with a short-term compounded annual growth rate of 366%.
- Financial data from The Company Check verifies an official revenue of ₹26.01 Lakh for the fiscal period ending March 31, 2024, reflecting a 136% growth rate compared to the prior year.

### Geographic Reach & Distribution Network — scored 81/100

> The company's local operating facility in Pune and basic nationwide online distribution via e-commerce sites were verified. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- According to primary research, the company relies on digital e-commerce platforms, specifically listing select drone battery and home inverter battery SKUs publicly for national retail distribution via Amazon India and industrial B2B listings [1.3.6, 1.3.7].

### Certifications & Compliance for Defense/Aerospace — scored 80/100

> General industrial certifications were partially verified through self-reported listings, but the absence of verifiable defense-specific credentials such as MIL-STD compliance or DGCA approval limits the company's standing. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Lixir Energy Private Limited self-reports holding an ISO 9001 certification alongside IEC61427-1 on its product distributions.
- According to primary research, the company lists custom products categorized as "Batteries for Military Drones" on commercial supplier databases, but lacks independent third-party verification or specific registration records for aerospace or defense procurement norms.

## Enervation Powertech Private Limited

Enervation Powertech · Trailblazers · X 85 · Y 75 · Overall 80

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 92/100

> The company displays a present but partial self-reported capability for supplying lower-grade drone batteries without certified or high-end industry performance metrics, leaving the evidence largely self-reported and unverified. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- According to primary research, product listings indicate that Enervation Powertech Private Limited manufactures basic custom drone lithium battery packs, such as an 11.1V 5000mAh Li-NMC battery with a self-reported 5C/10C discharge capability.
- Based on primary research, the company has not disclosed specific energy metrics (Wh/kg) or high-performance continuous power output for advanced UAV flight applications.

#### Cycle Life & Durability — scored 82/100

> The company states that its cells undergo precise grading to ensure pack reliability. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Enervation Powertech Private Limited self-reports having a 512-channel cell grading capability to check cell parameters like capacity, internal resistance, voltage, and cell consistency during assembly.
- Based on primary research, the company has not disclosed evidence regarding specific cycle life ratings, shelf life, or certified environmental test results for their drone-specific battery models.

#### Safety & Thermal Management — scored 82/100

> The company states its battery packs are safety-tested and feature basic protection circuits like thermal shutdown. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Enervation Powertech markets its customized drone batteries as "safety-tested" and highlights precision cell grading across parameters like internal resistance and cell consistency to promote pack reliability.
- General product aggregators note that the company deals with basic protection configurations, listing "thermal shutdown" among safety components.

#### Customization & Form Factor Flexibility — scored 94/100

> The company provides self-reported evidence of custom battery design and FPV drone compatibility, but lacks third-party verification for rapid prototyping or large-scale OEM customization. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Enervation Powertech Private Limited lists "Custom Battery Solutions" and drone batteries as part of its service segments, explicitly offering customized battery packs for drone applications.
- The company advertises an 11.1V 5000mAh Li-NMC 5C battery on B2B retail channels that is stated to be compatible with custom drone and FPV setups.

#### R&D & Innovation in Battery Chemistry — scored 77/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Enervation Powertech Private Limited manufactures generic standard lithium batteries, such as a "Drone Battery 11.1V 5000mAh Li-NMC 5C,". Based on primary research, the company has not disclosed proprietary chemical R&D, patents, or partnerships in advanced chemistries.

### Y — BUSINESS CAPABILITY

#### OEM Partnerships & Customer Contracts — scored 65/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

### Production Capacity & Scalability — scored 83/100

> The company is a small early-stage startup with under 10 employees and limited monthly assembly figures relative to mass industrial drone battery demand.

SUPPORTING POINTS

- Enervation Powertech Private Limited's corporate listing indicates they are a small-scale manufacturer operating with an employee headcount of up to 10 people.
- Product listings indicate localized assembly capabilities, noting a specific operational product capacity of "500 per month" for their localized custom battery configurations.

### Revenue Growth & Financial Health — scored 74/100

> The company operates at a limited financial scale, with small annual revenue of ₹68.5L and slightly negative growth, and lacks any notable venture capital funding or major public contracts demonstrating established industry strength. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- According to third-party data from Tracxn, Enervation Powertech Private Limited generated an operating revenue of ₹68.5 Lakhs (INR 6.85 million) for the financial year ending on March 31, 2025, with a year-over-year revenue growth rate (CAGR) of -4%.
- Financial registry profiles from Tofler reveal the company operates as a small unlisted entity with an authorized and paid-up share capital of just ₹1.00 Lakh (INR 100,000).

### Geographic Reach & Distribution Network — scored 65/100

> The company is a small-scale manufacturer operating solely from its head office/facility in Noida with a very localized business footprint, lacking verified evidence of a wider national or international distribution network. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Enervation Powertech Private Limited operates exclusively out of a single physical registered address located at D-203, Sector 10, Noida, Gautam Buddha Nagar, Uttar Pradesh, India.
- Based on primary research, the company has not disclosed evidence or records exist indicating third-party distributors, nationwide regional dealers, active export operations, or regional service centers specifically supporting their drone battery lineup.

### Certifications & Compliance for Defense/Aerospace — scored 89/100

> The company claims general ISO and BIS certifications relevant to commercial battery manufacturing. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Enervation Powertech Private Limited self-reports holding standard "ISO, BIS, CE, RoHS, UN38.3" certifications on its official website.

## Exodrone Systems Private Limited Vaayu ● Evolving Players X 82 Y 85 Overall 84

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 91/100

> The company provides specific capacity and C-rate discharge specifications for its drone batteries, a partial but documented capability, though explicit metrics on energy density (Wh/kg) remain self-reported or unstated.

SUPPORTING POINTS

- Exodrone Systems Private Limited supplies an indigenous line of drone batteries branded as "Vaayu," featuring a 6S Lithium-Ion configuration with a 24,000 mAh capacity and a continuous discharge rate of 15C.
- The company also provides 22,000 mAh Lithium Polymer (LiPo) batteries tailored for heavy-duty agricultural drone systems.

#### Cycle Life & Durability — scored 94/100

> The company shows self-reported evidence and compliance through its 400-cycle usage guarantee and BIS certification. Based on primary research, the company has not disclosed extensive third-party environmental and resilience test logs.

SUPPORTING POINTS

- Exodrone Systems Private Limited offers its indigenous "Vaayu" line of drone batteries (such as the 6S 24000mAh Li-Ion variant) featuring a manufacturer-guaranteed lifespan of up to 400 usage cycles.
- Product listings for the company's drone batteries indicate they are BIS Certified, demonstrating compliance with basic Indian regulatory safety and quality benchmarks.

#### Safety & Thermal Management — scored 65/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Exodrone Systems Private Limited manufactures a line of indigenous drone batteries named "Vaayu" (e.g., a 24000mAh 6S Li-Ion variant), which include a manufacturer's warranty and a performance guarantee of up to 400 cycles.

#### Customization & Form Factor Flexibility — scored 76/100

> The company supplies specialized drone battery lines but lacks verifiable evidence of providing custom battery geometries, tailorable electrical specifications, or rapid battery prototyping services. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Exodrone Systems Private Limited manufactures an indigenous drone battery line under the brand "Vaayu," offering specific standard configurations such as a 24000mAh 6S 15C Li-Ion battery.
- Based on primary research, the company has not disclosed further details.

#### R&D & Innovation in Battery Chemistry — scored 86/100

> The company has basic localized capability in assembling and offering commercial lithium-based drone battery packs. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Exodrone Systems Private Limited has introduced an indigenous line of drone batteries branded "Vaayu," featuring conventional Lithium-Ion chemistry (6S, 24000mAh) backed by a 400-cycle usage guarantee and a manufacturer's warranty.
- Public commercial listings indicate the company also supplies third-party or standard Lithium Polymer (LiPo) drone batteries (such as Herewin/Tattu 22000 mAh packs) alongside their own custom components.

### Y — BUSINESS CAPABILITY

#### OEM Partnerships & Customer Contracts — scored 80/100

> Secondary research confirms the existence of the indigenous Vaayu drone battery series marketed by the firm, but based on primary research, the company has not disclosed any named OEM partnerships, defense contracts, or specific long-term customer agreements for these batteries, given a lack of disclosed third-party data or corporate case studies.

SUPPORTING POINTS

- Exodrone Systems Private Limited lists its own "Vaayu" series 6S 24,000mAh lithium-ion battery on commercial marketplaces, but there are no third-party verified long-term supply agreements or named external drone manufacturers explicitly documented as OEM partners for these battery packs.
- Public corporate registers and profiles reveal the company acts primarily as an end-to-end drone manufacturer of custom UAV platforms—such as the Y3 Lance tactical VTOL and various agricultural sprayers—implying its battery systems are largely consumed by its own integrated aircraft line or sold as retail spares to individual end-users.

### Production Capacity & Scalability — scored 86/100

> Secondary research confirms the commercial availability of the Vaayu battery line and the small operational footprint of the firm, but based on primary research, the company has not disclosed specific macro manufacturing output figures (such as MWh/year capacity), precise assembly lines, or explicit factory floor expansions.

SUPPORTING POINTS

- According to primary research, product commercialization metrics listed on a B2B marketplace verify that Exodrone Systems manufactures and supplies its indigenous "Vaayu" 24,000 mAh Li-ion drone battery packs on a per-piece commercial catalog basis.
- Regulatory and operational filings available through corporate monitoring data on Tracxn and professional industry records list the company's organizational scale at 11 to 50 employees with an authorized share capital of ₹1 Lakh, establishing it as a small-scale enterprise or early-stage startup.

### Revenue Growth & Financial Health — scored 78/100

> Secondary research confirms the company's financial status as an early-stage bootstrapping entity generating revenue below ₹10 Cr, while based on primary research, the company has not disclosed details on net profitability.

SUPPORTING POINTS

- Exodrone Systems Private Limited generated less than ₹10 Cr in revenue for the financial year ending on March 31, 2025, according to corporate tracking data published by Tracxn.
- The company operates as an unfunded entity that has not raised external venture capital or formal government grant financing since its incorporation in December 2020, as documented in its institutional Tracxn Profile.

### Geographic Reach & Distribution Network — scored 92/100

> Secondary research confirms the company's local manufacturing presence in Hyderabad and direct digital distribution footprint in India via wholesale platforms, but based on primary research, the company has not disclosed third-party evidence of brick-and-mortar dealer networks, regional service centers, or specific export agreements, limiting verified reach primarily to self-managed domestic distribution.

SUPPORTING POINTS

- Exodrone Systems Private Limited has a registered office at Dilsukhnagar and corporate/manufacturing facilities at Hasthinapuram and Saroornagar in Hyderabad, Telangana, and is categorized as a drone accessory distributor/dealer locally on Justdial.
- According to primary research, the company maintains an active digital storefront profile to distribute its indigenous 24000mAh "Vaayu" series lithium-ion drone batteries directly from its Hyderabad base to buyers across India.

### Certifications & Compliance for Defense/Aerospace — scored 91/100

> Bureau of Indian Standards product certification and self-reported alignment with Indian DGCA regulations were verified, but based on primary research, the company has not disclosed official third-party defense audit listings, MIL-STD testing certifications, or direct validation of compliance with defense procurement norms.

SUPPORTING POINTS

- According to primary research, Exodrone Systems states that its proprietary Vaayu lithium-ion batteries feature BIS Certified Product compliance as detailed in its distribution catalog.
- The company indicates that its system designs align with DGCA regulations for customers in India and targets tactical defense applications on its official services portal.

## RCRS Innovations Limited EXEGI ● Evolving Players X 82 Y 83 Overall 82

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 68/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. RCRS Innovations Limited lacks verifiable public specifications, technical data sheets, or third-party documentation on Wh/kg or C-rate outputs optimized for unmanned aerial vehicles.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

#### Cycle Life & Durability — scored 82/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- RCRS Innovations Limited produces lithium-ion and LiFePO4 battery packs under its EXEGI brand, listing Aerospace & Defence as a targeted application category, but its public specification sheets only provide broad marketing claims of industry-leading performance and durability without concrete metrics.
- General technical literature on the company's site lists a broad chemistry ceiling of up to 2000 cycles for general LiFePO4 configurations and notes that their manufacturing process uses automated cell sorting and optimized thermal management for longer operational life, but does not provide specific cycle life numbers, vibration tolerances, or environmental test results engineered for unmanned aerial vehicles (UAVs).

#### Safety & Thermal Management — scored 82/100

> Although RCRS Innovations lists UAV applications for its battery packs, its technical claims and safety certifications remain self-reported at a general level rather than verified by drone-specific third-party credentials like UN38.3 or IEC 62133.

SUPPORTING POINTS

- RCRS Innovations Limited utilizes integrated Battery Management Systems (BMS) with diagnostics and proprietary designs to prevent thermal runaway and protect against external impact or crush.
- The company reports testing to IEEE 1625 and 1725 battery standards, uses BIS-approved raw materials, and holds ISO 9001:2015 and CE certifications.

#### Customization & Form Factor Flexibility — scored 97/100

> The company shows partial, self-reported evidence of offering customizable lithium-ion configurations for specialized fields like UAVs and defense.

SUPPORTING POINTS

- RCRS Innovations Limited lists aerospace and defense batteries as an official corporate specialty, featuring self-reported design capabilities for custom configurations including uncrewed aerial vehicles.
- The company maintains an in-house engineering team tasked with optimizing cell combinations to tailor battery sizing, dimensions, and voltage configurations to client specifications.

#### R&D & Innovation in Battery Chemistry — scored 82/100

> While the company demonstrates standard pack-level engineering capabilities, the absence of publicly verifiable R&D partnerships, patents, or next-generation chemistry announcements leaves this capability weak and unproven. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- RCRS Innovations Limited maintains an in-house R&D team focusing on custom structural pack designs, automated cell sorting, and thermal management for applications including aerospace and defense.
- The company's published chemical literature focuses exclusively on standard commercial chemistries like LCO, LFP, and NMC, with no public mention of advanced or proprietary next-generation developments such as solid-state or lithium-sulfur chemistries.

### Y — BUSINESS CAPABILITY

#### OEM Partnerships & Customer Contracts — scored 76/100

> The company states it can supply custom battery solutions for UAV and defense applications, but there is an absence of verifiable third-party contracts or named OEM partnerships specific to the drone industry. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- According to primary research, RCRS Innovations Limited self-reports capability in designing customized aerospace and defence battery packs, specifically listing "unmanned aerial vehicles (UAV)" under its targeted applications via its business profile.
- Based on primary research, the company has not disclosed named drone manufacturer or defense customer contract, contract value, or UAV case study.

### Production Capacity & Scalability — scored 93/100

> The company has verifiable operational lithium-ion manufacturing capabilities reaching 250 MWh/year, though specific output data for UAV-engineered applications remains absent from public filings.

SUPPORTING POINTS

- The company lists Aerospace & Defence batteries as part of its specialties on corporate profiles, but its primary manufacturing output and scale-up activities target bulk clean energy storage, BESS, and solar lighting solutions.

### Revenue Growth & Financial Health — scored 78/100

> While the company shows clear corporate-level financial growth within the general energy storage space, there is minimal verified evidence linking its revenue explicitly to high-performance drone and UAV power management systems.

SUPPORTING POINTS

- RCRS Innovations Limited demonstrated corporate-wide financial performance, growing its total operating revenue from ₹115.85 crore in FY24 to ₹132.76 crore in FY25, alongside filing for an IPO on the NSE Emerge platform.
- The company operates primarily as a manufacturer of customized lithium-ion battery packs and solar PV modules under the brand "EXEGI", focused on solar lighting and general clean energy storage applications, rather than displaying dedicated revenue segmentation for the drone/UAV battery sector.

### Geographic Reach & Distribution Network — scored 79/100

> The company's geographic footprint and distribution network within the niche drone battery segment are heavily unproven, relying primarily on generalized, self-reported B2B capabilities from its broader product lines. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- RCRS Innovations Limited operates two manufacturing facilities in the Delhi-NCR region covering over 50,000 sq ft, with its primary corporate offices located in Noida and East Delhi.
- While the company lists aerospace and defense batteries among its customization capabilities and maintains self-reported supply registrations, it lacks third-party verified data, lists of global countries served, or established distribution agreements specifically for the drone battery vertical.

### Certifications & Compliance for Defense/Aerospace — scored 90/100

> While RCRS Innovations is ISO 9001:2015 certified and claims engineering competency for defense/aerospace battery enclosures, based on primary research, the company has not disclosed it holding specific aerospace-grade or DGCA type certifications tailored for UAVs, keeping its evidence level partial and self-reported.

SUPPORTING POINTS

- RCRS Innovations Limited holds over 20 quality and safety certifications, including ISO 9001:2015 (Quality Management Systems), ISO 14001:2015, CE marking, RoHS, and BIS approval.

## Lithium Energy Experts Pvt. Ltd. LEEPL ● Evolving Players X 83 Y 79 Overall 81

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 91/100

> While basic voltage and capacity metrics are self-reported via standard B2B product catalogs, specific technical indicators of performance such as Wh/kg energy density figures, continuous power outputs, and based on primary research, the company has not disclosed C-rate discharge specifications.

SUPPORTING POINTS

- According to primary research, the company offers a "LEEPL 30 Ah Lithium Ion Drones Camera Battery" specified at a capacity of 30 Ah and an exact voltage profile of 11.1 V under its brand name LEEPL, as detailed in its commercial catalog listing.
- The manufacturer publicly claims its range of lithium battery packs optimized for drone applications spans across dual cell chemistries—namely Lithium Polymer (LiPo) and high-voltage Lithium Polymer (LiHV)—as stated on its official Company Product Portfolio.

#### Cycle Life & Durability — scored 82/100

> Secondary research confirms basic e-commerce product categories and general marketing claims, but based on primary research, the company has not disclosed exact cycle life ratings, shelf life, or resilience to vibration, humidity, and temperature swings, given the absence of public laboratory data.

SUPPORTING POINTS

- According to primary research, a product listing portfolio demonstrating that the company manufactures general configurations such as 11.1V 18Ah Lithium-Ion and 12.8V LFP battery packs, but provides no specific cycle life parameters or technical datasheets explicitly tailored for UAV flight demands.
- According to primary research, self-reported corporate claims compiled via its business listing asserting generalized product benefits without any accompanying third-party verification or documented testing metrics.

#### Safety & Thermal Management — scored 65/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Company product catalog details that Lithium Energy Experts Pvt. Ltd. manufactures standard commercial lithium-ion battery packs alongside generalized lithium-polymer options.
- Public corporate profiles and regulatory listings note that the company focuses its safety claims broadly on stability for generalized consumer lines rather than specialized UAV-certified testing.

#### Customization & Form Factor Flexibility — scored 94/100

> Secondary research confirms the company's self-reported focus on application-specific customization, flexible battery dimensions, and varying chemistry combinations (LiPo/LiHV), but based on primary research, the company has not disclosed third-party validation, named industrial client projects, or specific rapid-prototyping case studies.

SUPPORTING POINTS

- According to primary research, the company showcases the ability to supply tailored dimensional configurations, stating that their drone camera lithium-ion battery packs are engineered with "sharper flexibility" and lower weight specifically for varying robotic and quadcopter applications, as documented in its product listing (LEEPL 30 Ah Product Listing).

#### R&D & Innovation in Battery Chemistry — scored 84/100

> Secondary research confirms the use of standard commercial chemistries through catalog data. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- The company self-reports providing standard Lithium Polymer (LiPo) and Lithium Polymer High Voltage (LiHV) battery packs on its official listings.
- Based on primary research, the company has not disclosed patents, R&D facility certifications, or next-generation chemistry projects.

### Y — BUSINESS CAPABILITY

#### OEM Partnerships & Customer Contracts — scored 69/100

> Based on primary research, the company has not disclosed any named drone manufacturers, defense organization clients, contract values, or case studies, since public records show exclusively general, small-scale component manufacturing without public OEM integrations.

SUPPORTING POINTS

- According to primary research, corporate profile listings verify that Lithium Energy Experts Pvt. Ltd. acts as a basic component supplier selling generic lithium-ion/polymer packs and medical ventilator batteries rather than participating in public OEM drone partnerships.
- Financial and legal tracking registries from Tracxn show the company operates as an unfunded local entity with zero recorded defense contracts, multi-year OEM supply agreements, or enterprise customer case studies.

### Production Capacity & Scalability — scored 85/100

> While the physical facility location and basic corporate size metrics were verified, explicit manufacturing output metrics (such as MWh/year), automated production line details, or concrete expansion plans could not be confirmed through secondary research. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- According to the company's official corporate profile, Lithium Energy Experts Pvt. Ltd. operates its physical manufacturing operations out of a single facility located at F-118, First Floor, Sector-8, Noida, Uttar Pradesh, India.
- Publicly available corporate data and business profiles show that the company has a small operational scale, maintaining a workforce of approximately 11 to 25 employees and an annual turnover of ₹5–25 Crore.

### Revenue Growth & Financial Health — scored 87/100

> Financial parameters concerning statutory capital size, operational revenue range, and year-over-year growth were verified through registry and platform data, but based on primary research, the company has not disclosed institutional venture capital funding or government grants.

SUPPORTING POINTS

- Corporate registry filings up to March 31, 2025, indicate that Lithium Energy Experts Private Limited generated less than ₹10 Crore in operating revenue, supported by an authorized and paid-up capital of ₹1 Lakh as recorded on Tracxn.
- According to primary research, data updated through mid-2026 on Tracxn notes a 32% year-over-year revenue growth rate, while self-reported revenue brackets between ₹5 Crore and ₹25 Crore are detailed on its business listing.

### Geographic Reach & Distribution Network — scored 89/100

> Secondary research confirms the company's single physical facility and general self-reported export capacity, but based on primary research, the company has not disclosed specific countries served, regional service centers, or formal distribution agreements.

SUPPORTING POINTS

- The company operates a localized manufacturing facility and corporate office at F-118, Sector-8, Noida, Uttar Pradesh, India, as verified by official GST registration records.
- According to primary research, the company self-reports an export percentage of up to 20% and domestic operations supplying drone camera battery packs across India, according to the company's business profile.

### Certifications & Compliance for Defense/Aerospace — scored 65/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. Based on primary research, the company has not disclosed any holding of DGCA, MIL-STD, ISO 9001, or Indian defense procurement compliance for drone battery packs, given the absence of documented defense or aerospace certifications.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed information for this parameter. No defense or aerospace credentials could be found in public registries or project listings, as the company's verified portfolio focuses primarily on standard commercial applications such as solar street lights and medical ventilator batteries.
- Based on primary research, the company has not disclosed information for this parameter. No third-party certificates or regulatory listings for specialized military or aviation frameworks (such as DGCA type certifications or MIL-STD testing compliance) are documented for the company's products.

## Aqueouss ● Evolving Players

X 72 Y 86 Overall 79

### X — PRODUCT CAPABILITY

#### Energy Density & Power Output — scored 68/100

> The company mentions marketing drones as a target application on its website but lacks any verifiable or detailed technical data on energy density and discharge rates for the drone segment. Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Not available

#### Cycle Life & Durability — scored 68/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. Based on primary research, the company has not disclosed the cycle life, environmental test results, or durability specs of Aqueouss energy storage units specifically engineered for the drone market.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

#### Safety & Thermal Management — scored 66/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score. Aqueouss makes generic claims regarding its manufacturing quality, specifying "thermal protection and safety design,".

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

#### Customization & Form Factor Flexibility — scored 94/100

> While the company presents self-reported capabilities for custom lithium battery design and prototyping that apply to the drone vertical, it lacks extensive third-party verification or granular drone-specific form-factor records.

SUPPORTING POINTS

- Aqueouss maintains a structured custom workflow consisting of customer requirements analysis, battery design and development, and prototype testing for several applications, explicitly including drones.
- The company actively promotes self-reported "customised lithium battery pack solutions" tailored to various products and mission dimensions.

#### R&D & Innovation in Battery Chemistry — scored 65/100

> Based on primary research, the company has not disclosed information for this parameter. The available evidence is insufficient to substantiate this parameter or to justify the assigned score.

SUPPORTING POINTS

- Based on primary research, the company has not disclosed supporting evidence.

### Y — BUSINESS CAPABILITY

#### OEM Partnerships & Customer Contracts — scored 82/100

> Based on primary research, the company has not disclosed further details.

SUPPORTING POINTS

- Aqueouss reports on its official channels that its custom lithium-ion and LiFePO4 battery packs are utilized across various industries, explicitly listing drones and robotics alongside solar and defense applications.
- Public procurement records from the Government of India e-Marketplace confirm that Aqueouss secures direct government supply awards for its lithium battery packs, such as contract GEMC-511687730456222 issued in Maharashtra.

### Production Capacity & Scalability — scored 94/100

> Secondary research confirms that the company maintains an in-house assembly and production plant in New Delhi, but based on primary research, the company has not disclosed exact annual manufacturing output (MWh/year), discrete production line counts, and forward-looking scalability or expansion plans, given the lack of disclosed third-party operational audits.

SUPPORTING POINTS

- Aqueouss self-reports operating a domestic battery pack manufacturing facility with advanced equipment located in the Mayapuri Industrial Area, New Delhi, India.
- Public corporate registries and business profiles confirm Aqueouss operates as a small-to-medium enterprise (SME) with an estimated company size of 11 to 50 employees.

### Revenue Growth & Financial Health — scored 81/100

> General revenue brackets were verified through business registries, but based on primary research, the company has not disclosed standalone venture capital funding, government grants, and specific revenue growth for drone batteries.

SUPPORTING POINTS

- Official operational profile data shows Aqueouss operates as a New Delhi-based partnership firm generating an annual turnover estimated between ₹2.5 Crore to ₹5 Crore.
- Media profiles report the company clocking ₹3.8 Crore in annual revenue across generic industrial and mobility applications rather than isolated financial tracking for UAVs.

### Geographic Reach & Distribution Network — scored 84/100

> Secondary research confirms the company's primary corporate facilities, domestic supply chain nodes, and broad export baseline, but was unable to identify any third-party validated regional distributors or dedicated domestic/export dealer agreements tailored specifically to its drone battery vertical.

SUPPORTING POINTS

- Aqueouss maintains a primary corporate office and warehouse facility spanning approximately 2,000 square meters in the Mayapuri Industrial Area Phase II, Delhi, alongside an established third-party managed regional logistics hub via Goldfishh SupplyChain Technologies in Hyderabad, Telangana.
- According to the company's verified commercial distribution profile, its global exposure is limited to minor trading actions with a stated export threshold capped at up to 20% of its business volume, while its dedicated regional dealership program is explicitly directed at domestic inverter, ESS, and e-rickshaw battery distribution rather than dedicated drone infrastructure channels.

### Certifications & Compliance for Defense/Aerospace — scored 90/100

> While secondary research confirms that Aqueouss lists "Defence" as a target industry and provides generic NABL/IEC lab test reports via procurement platforms, based on primary research, the company has not disclosed third-party evidence of specific aerospace-grade or military-grade hardware certifications such as DGCA type certifications or MIL-STD compliance, leaving this a partial capability.

SUPPORTING POINTS

- Aqueouss publicly states that its custom lithium-ion and LiFePO4 battery packs are tailored for multiple industrial sectors including robotics, railways, and defense platforms.
- Government procurement records on the Government e-Marketplace (GeM) confirm that Aqueouss submits IEC-compliant lithium-ion solar batteries with type test reports available from Central Govt/NABL/ILAC accredited laboratories upon buyer demand.

## Parameter Definitions

### X — Product Capability

1. **Energy Density & Flight Time**
   Measures the battery's energy per unit weight (Wh/kg) and its ability to maximize drone flight time per charge. Evidence includes published specs, third-party test results, and customer benchmarks showing flight duration improvements.
2. **Battery Chemistry & Safety**
   Assesses the safety and stability of the battery chemistry (e.g., LiPo, Li-ion, solid-state) under Indian operating conditions like high temperatures. Evidence includes safety certifications, thermal runaway test reports, and failure rate data.
3. **Customization & Form Factor Flexibility**
   Evaluates the ability to design batteries in various shapes, sizes, and voltages to fit different drone models (e.g., agricultural, surveillance, delivery). Evidence includes custom battery designs for specific OEMs and rapid prototyping capabilities.
4. **Cycle Life & Durability**
   Refers to the number of charge-discharge cycles the battery can withstand while maintaining performance, and its resilience to vibration and environmental stress. Evidence includes cycle life test data and field reliability reports from Indian drone operators.
5. **Charging Speed & Smart BMS**
   Covers fast-charging capability and the sophistication of the battery management system (BMS) for monitoring, balancing, and extending battery life. Evidence includes charging time specs, BMS features (e.g., cell balancing, state-of-health estimation), and compatibility with smart chargers.

### Y — Business Capability

1. **OEM Partnerships & Contract Wins**
   Tracks the number and significance of partnerships with Indian drone manufacturers (OEMs) and contracts with government or private entities. Evidence includes named OEM customers, contract values, and letters of intent.
2. **Revenue & Growth Rate**
   Measures the company's annual revenue from drone battery sales in India and its year-over-year growth percentage. Evidence includes audited financial statements, investor reports, and public disclosures.
3. **Manufacturing Capacity & Localization**
   Assesses the scale of manufacturing operations in India, including production capacity (MWh/year) and level of local sourcing to meet 'Make in India' norms. Evidence includes factory locations, capacity expansion announcements, and local content percentages.
4. **Distribution & After-Sales Network**
   Evaluates the breadth of the distribution network across Indian states and the quality of after-sales support (e.g., warranty, replacement, technical support). Evidence includes number of distributors, service centers, and customer satisfaction metrics.
5. **Funding & Strategic Investments**
   Refers to the amount and source of funding raised (equity, debt, grants) and strategic investments from Indian or global players. Evidence includes funding rounds, investors, and government grants under schemes like PLI.

## Research Methodology

This quadrant was compiled independently through structured desk research, supplemented by primary research where available. It is not based on a survey or on vendor self-submission. The steps below describe how companies were identified, screened, evidenced and scored.

**1 Market Definition**

The market scope and the type of company that qualifies were defined first. Five parameters were then set for each axis, each with a written definition: the X axis measures core capability (products, services or technology) and the Y axis measures business capability.

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**2 Desk and Primary Research**

Desk research covered company websites, filings, trade directories, industry publications and news. Primary research drew on B2B marketplace listings, where available, to confirm company offerings. Coverage was extended region by region until no new qualifying companies emerged.

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**3 Screening**

Each candidate was screened against the market's inclusion criteria and required commercial role, and its market activity was verified. Distributors, resellers, consultancies, contract manufacturers and media names were excluded, and duplicate entries were consolidated.

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**4 Evidence Compilation**

Every screened company was assessed on all ten parameters. For companies profiled on the chart, evidence was recorded parameter by parameter. Findings from B2B marketplace listings are reported as primary research; items not verified through primary research are marked as such.

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**5 Scoring**

Each parameter is rated on a 0–100 scale against its written definition, based on the evidence available for that company. A company's X and Y scores are the simple average of its five parameter ratings on each axis, and its overall score is the average of X and Y.

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**6 Normalization & Selection**

X and Y scores are rescaled to a 65–100 range across all companies assessed, preserving their rank order. The chart shows an equal number of companies from each of the four quadrants, with boundaries set at the median. All other companies are listed in the table below the chart.

Scoring inputs: each axis is built from five market-specific parameters, defined in the Parameter Definitions section. All parameters are equally weighted.